IBM Developer
SKILLS NETWORK

IBM APPLIED DATA SCIENCE CAPSTONE

SpaceX Falcon 9 First-Stage Landing Prediction

Winning Space Race with Data Science

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October 18, 2026

github.com/treinart/IBM-Data-Science-Professional-Certificate

AVAILABLE FORMATS

PowerPoint Deck PDF Presentation Google Slides



Outline

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Executive Summary

Key Finding: Landing success improved on later flights. The final model identified every success, but only half of failures.

	METHODOLOGY USED	SUMMARY OF RESULTS
01	**API collection, web scraping, outcome labeling**	90-launch model set: 60 successful landings, 30 unsuccessful outcomes.
02	**Visual EDA by flight, payload, orbit, site and year**	Later flights improved. 2019: 9/10; 2020: 16/19. VLEO: 12/14; GTO: 14/27.
03	**SQL queries on sites, payloads, missions and landings**	Four sites; NASA (CRS) payload totaled 45,596 kg; first ground-pad success: Dec 22, 2015.
04	**Folium mapping of outcomes and nearby features**	Four sites mapped in California and Florida; launch clusters and proximity distances measured.
05	**Dash site and payload filters with pie and scatter views**	Dashboard sample: 24 successes across sites; 2,000–6,000 kg range: 18/34 successes.

PREDICTIVE MODELING

4 tuned classifiers

One-hot encoding, scaling, 72/18 split and 10-fold grid search.

SELECTED SVM

83.3%

15/18 test • LR + KNN tied

84.8% cross-validation

FINAL CLASS RESULTS

Success (12): precision .80 | recall 1.00 | F1 .89

Failure (6): precision 1.00 | recall .50 | F1 .67

3 false positives • 0 false negatives

Introduction

Why do some Falcon 9 first stages land while others do not?

THE WORK BEHIND THIS DECK

I worked through the launch records to see whether flight history, payload, orbit, and launch site had anything to do with the landing result.

I checked the records with charts and SQL, mapped the launch sites, built a dashboard, and tested four prediction models.

This presentation shows what I found and where the models still missed.

WHAT I WANTED TO FIND OUT

01

Did successful landings become more common as Falcon 9 flew more missions?

02

Did outcomes differ by payload, orbit, booster, or launch site?

03

What did the SQL results and maps reveal that the charts alone could not?

04

How well could the models identify both successful and unsuccessful landings?

Section 1

Methodology



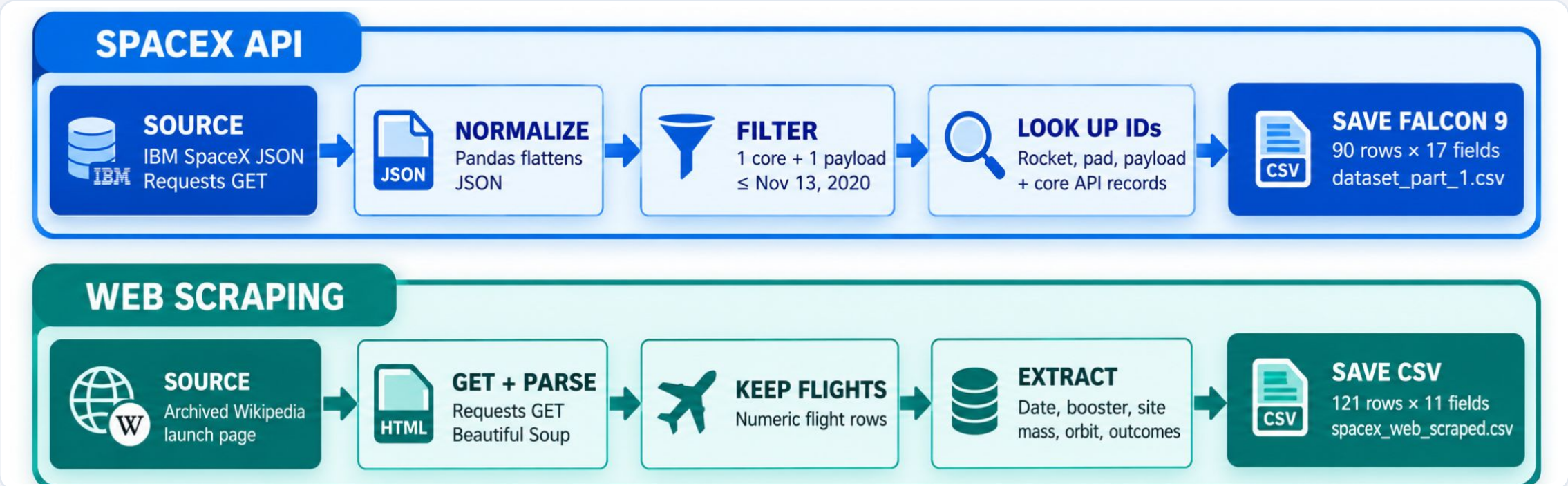
Methodology Executive Summary

Structured 5-stage data science pipeline from raw flight records to tested landing models

STAGE / PHASE	METHODOLOGY & KEY ACTIVITIES	TOOLS & LIBRARIES
01 COLLECT	Fetched SpaceX launch records via REST API and scraped archived Falcon 9 launch tables from Wikipedia.	Requests, Beautiful Soup
02 WRANGLE	Filtered Falcon 9 flights, imputed missing payload mass by mean, and encoded binary landing outcomes (0/1).	Pandas, NumPy
03 EDA + SQL	Analyzed flight correlations by launch site, orbit, and payload mass using statistical queries and visual plots.	Matplotlib, Seaborn, SQLite
04 MAP + DASH	Built interactive maps of launch sites, landing outcomes, and coastal distances alongside real-time payload filters.	Folium, Plotly, Dash
05 PREDICT	Standardized features, tuned 4 classifiers (Logistic Regression, SVM, Decision Tree, KNN) via 10-fold CV, and evaluated test accuracy.	scikit-learn

How the Launch Datasets Were Collected

Structured dual-pipeline strategy combining official REST API retrieval and HTML table scraping



SPACEX API PIPELINE

Notebook: jupyter-labs-spacex-data-collection-api-v2.ipynb

Output File: dataset_part_1.csv (90 rows × 17 fields)

WEB SCRAPING PIPELINE

Notebook: jupyter-labs-webscraping.ipynb

Output File: spacex_web_scraped.csv (121 rows × 11 fields)

Data Collection – SpaceX API

KEY PHRASES

PRIMARY CALL

GET /v4/launches/past

RECORD PREP

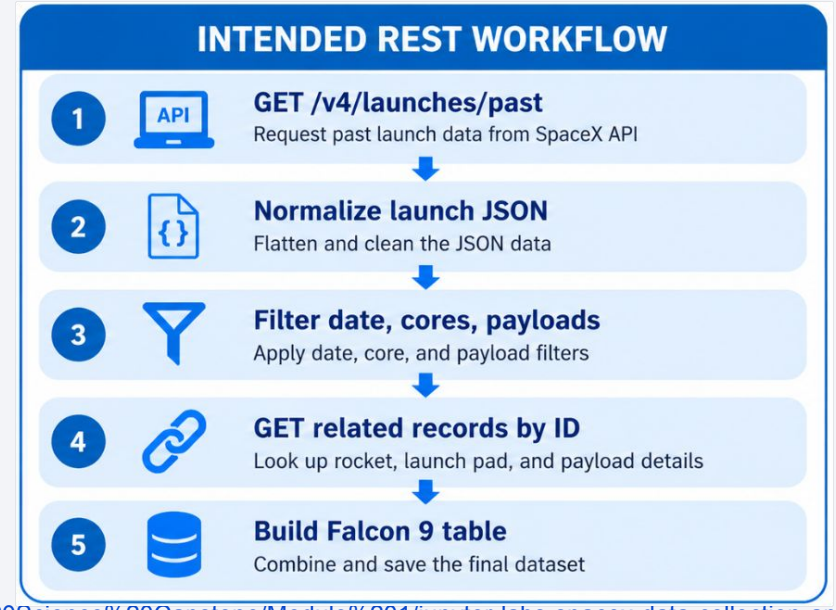
Normalize JSON; one core and one payload

LINKED LOOKUPS

Rocket, launchpad, payload, and core IDs

TARGET OUTPUT

Falcon 9 launch table → dataset_part_1.csv



https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%201/jupyter-labs-spacex-data-collection-api-v2.ipynb

SpaceX API Availability Note

Checked: Multiple times on October 17 and again on October 18, 2026.

This notebook retains the original IBM lab code and its required SpaceX v4 API requests. At the time of testing, `api.spacexdata.com` could not be resolved. Requests to the launch, rocket, payload, launchpad, and core endpoints therefore returned no JSON response.

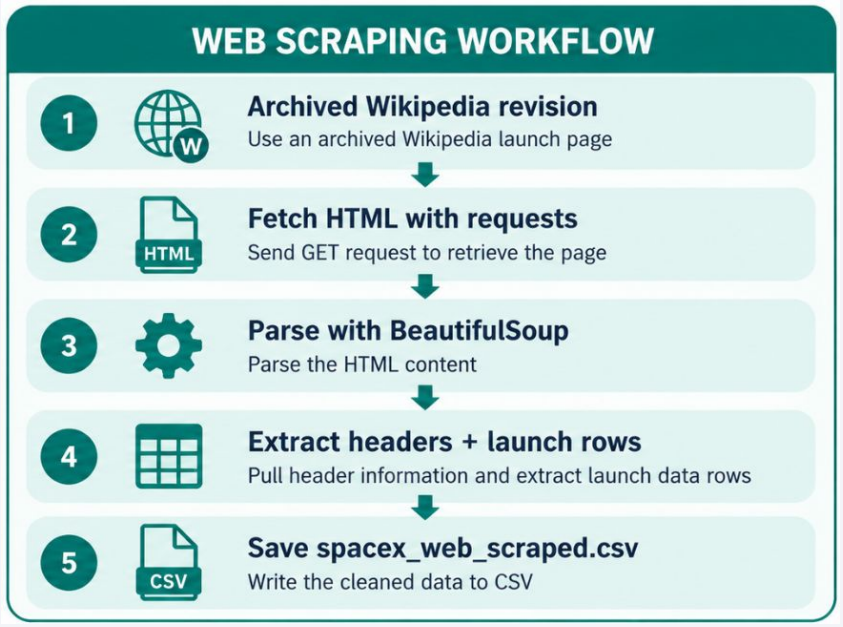
This is an external API availability issue, not an error in the notebook logic. The notebook intentionally preserves the original API calls and their resulting error rather than substituting a different data source or changing the course exercise.

I also checked for a compatible historical JSON replacement but did not find a source that preserved the exact fields and relationships required by this lab. To continue the downstream course labs, I used IBM's published course dataset:

https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/dataset_part_1.csv

That CSV is the IBM-provided input used by the following data-wrangling lab. It supports the later notebooks, but it does not replace or hide the original API collection failure documented here.

Data Collection – Web Scraping



KEY PHRASES

SOURCE PAGE
Archived Falcon 9 and Falcon Heavy launches

REQUEST + PARSE
requests.get() and BeautifulSoup

TABLE EXTRACTION
Read launch table headers and rows

SAVED RESULT
121 records · 11 fields · CSV export

https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%201/jupyter-labs-webscraping.ipynb

NOTEBOOK OUTPUT · FIRST FIVE SCRAPED LAUNCHES

	Flight No.	Launch site	Payload	Payload mass	Orbit	Customer	Launch outcome	Version Booster	Booster landing	Date	Time
0	1	CCAFS	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	F9 v1.07B0003.1	Failure	4 June 2010	18:45
1	2	CCAFS	Dragon	0	LEO	NASA	Success	F9 v1.07B0004.1	Failure	8 December 2010	15:43
2	3	CCAFS	Dragon	525 kg	LEO	NASA	Success	F9 v1.07B0005.1	No	22 May 2012	07:44
3	4	CCAFS	SpaceX CRS-1	4,700 kg	LEO	NASA	Success	F9 v1.07B0006.1	No attempt	8 October 2012	00:35
4	5	CCAFS	SpaceX CRS-2	4,877 kg	LEO	NASA	Success	F9 v1.07B0007.1	No	1 March 2013	15:10

Data Wrangling

KEY PHRASES

INPUT

dataset_part_1.csv · 90 Falcon 9 launches

DATA CHECKS

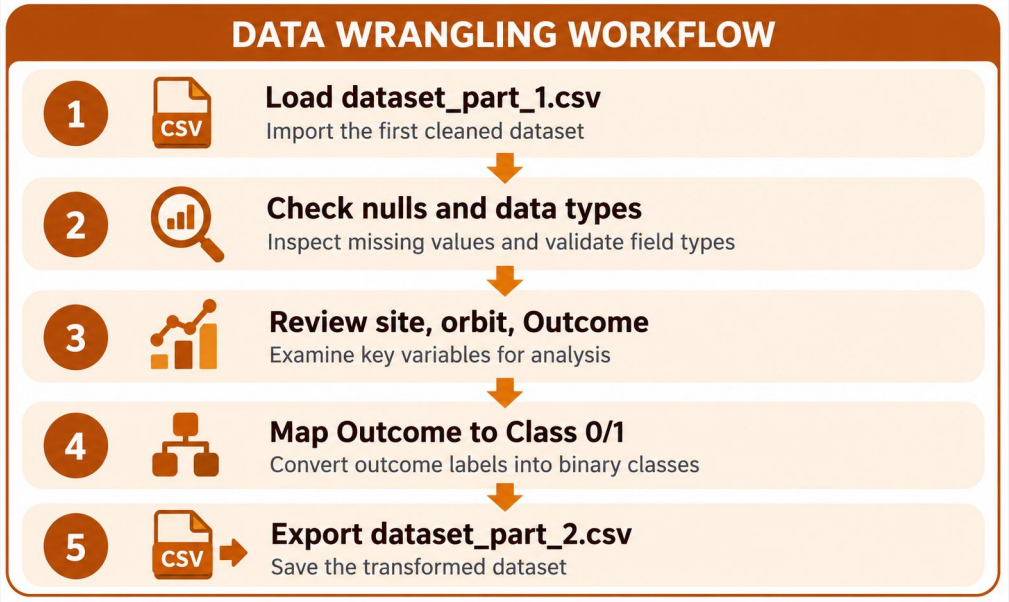
Missing values, data types, site and orbit counts

LANDING TARGET

Outcome → Class: 1 success, 0 otherwise

OUTPUT

dataset_part_2.csv · 90 rows, 18 fields



https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%201/labs-jupyter-spacex-Data%20wrangling-v2.ipynb

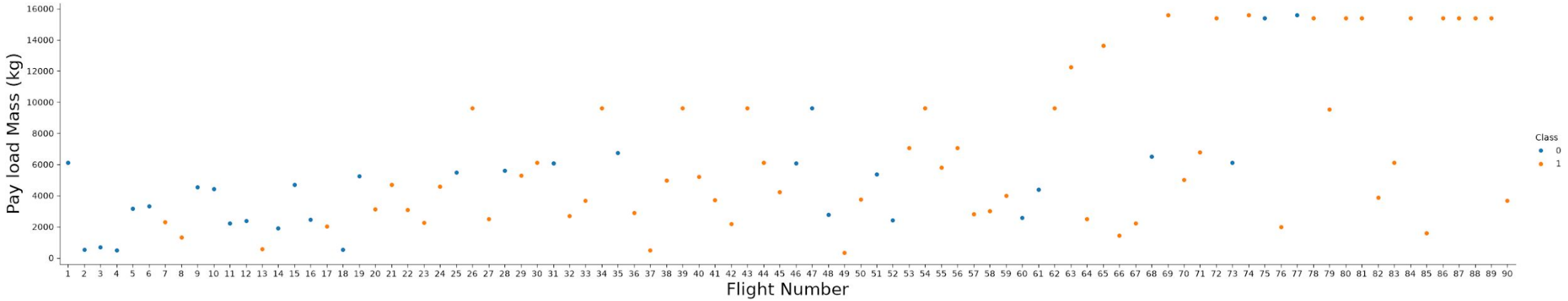
NOTEBOOK OUTPUT · FIRST TEN WRANGLED LAUNCHES

	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial	Longitude	Latitude	Class
0	1	2010-06-04	Falcon 9	6104.959412	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0003	-80.577366	28.561857	0
1	2	2012-05-22	Falcon 9	525.000000	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0005	-80.577366	28.561857	0
2	3	2013-03-01	Falcon 9	677.000000	ISS	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0007	-80.577366	28.561857	0
3	4	2013-09-29	Falcon 9	500.000000	PO	VAFB SLC 4E	False Ocean	1	False	False	False	NaN	1.0	0	B1003	-120.610829	34.632093	0
4	5	2013-12-03	Falcon 9	3170.000000	GTO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B1004	-80.577366	28.561857	0
5	6	2014-01-06	Falcon 9	3325.000000	GTO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B1005	-80.577366	28.561857	0
6	7	2014-04-18	Falcon 9	2296.000000	ISS	CCAFS SLC 40	True Ocean	1	False	False	True	NaN	1.0	0	B1006	-80.577366	28.561857	1
7	8	2014-07-14	Falcon 9	1316.000000	LEO	CCAFS SLC 40	True Ocean	1	False	False	True	NaN	1.0	0	B1007	-80.577366	28.561857	1
8	9	2014-08-05	Falcon 9	4535.000000	GTO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B1008	-80.577366	28.561857	0
9	10	2014-09-07	Falcon 9	4428.000000	GTO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B1011	-80.577366	28.561857	0

EDA with Data Visualization

CHART TITLE	CHART TYPE	REASON FOR USING IT
Flight Number vs. Payload Mass	Scatter plot	Keeps each launch visible across flight order and payload weight; grouping the data would hide individual landing outcomes.
Flight Number by Launch Site	Scatter plot	Separates each site's flight history so the overall trend does not stand in for every site.
Payload Mass by Launch Site	Scatter plot	Shows the payload range at each site; an average would conceal which weights were never flown there.
Average Landing Success Rate by Orbit	Bar chart	Orbit types are separate categories, so bars put their success rates on a common scale.
Flight Number by Orbit Type	Scatter plot	Shows whether the pattern across later flights holds within each orbit type.
Payload Mass by Orbit Type	Scatter plot	Keeps the payload ranges of different orbit types separate when reading landing outcomes.
Average Landing Success Rate by Year	Line chart	Years have an order; the line shows changes and reversals in the annual rate.

SAMPLE OUTPUT · FLIGHT NUMBER VS. PAYLOAD MASS



https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%202/jupyter-labs-eda-dataviz-v2.ipynb

EDA with SQL

QUERIES 1–5

01 List distinct launch sites with `SELECT DISTINCT`.

02 Return five launch records from sites beginning CCA with `LIKE` and `LIMIT`.

03 Total NASA (CRS) payload mass with `SUM`.

04 Calculate average F9 v1.1 payload mass with `AVG`.

05 Find the first successful ground-pad landing date with `MIN`.

QUERIES 6–10

06 List boosters that landed on a drone ship with payloads above 4,000 and below 6,000 kg.

07 Count each mission outcome with `GROUP BY` and `COUNT`.

08 Find every booster tied for the maximum payload mass with `MAX`.

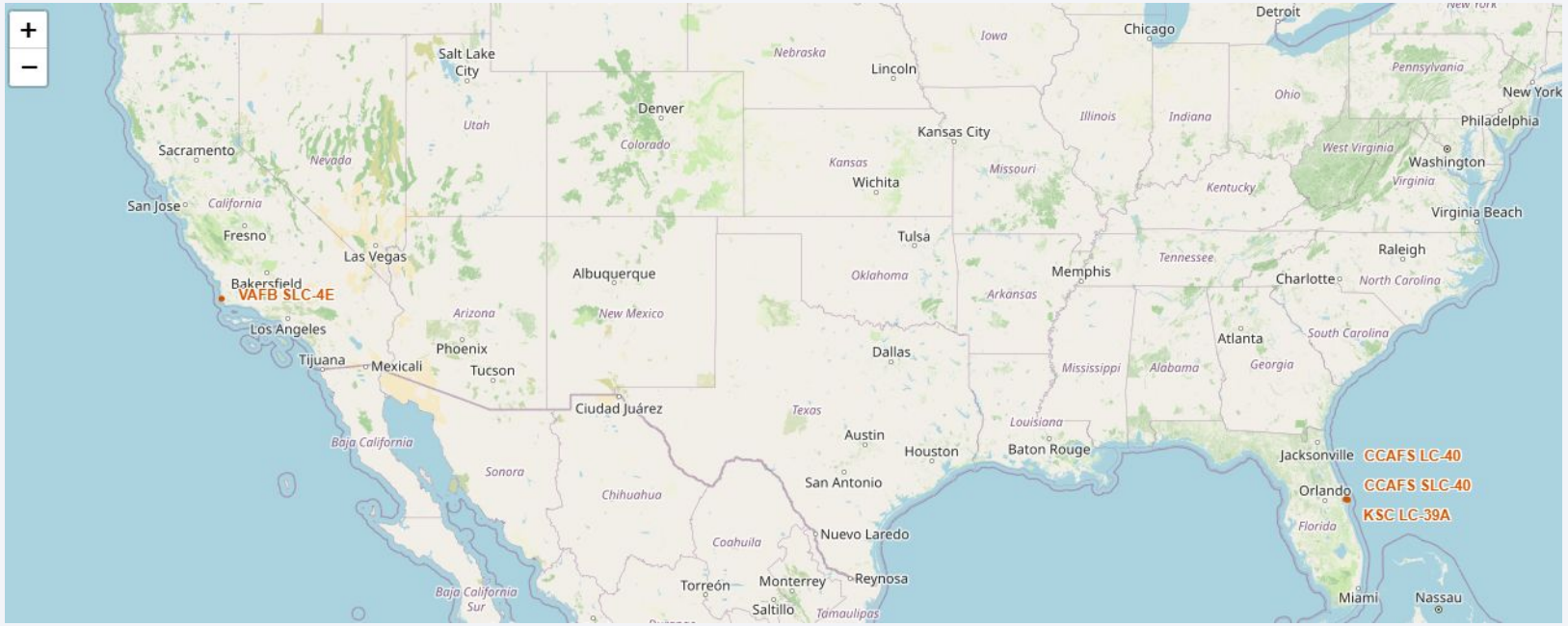
09 Extract the month, booster and launch site for 2015 drone-ship landing failures.

10 Count and rank landing outcomes from June 4, 2010 through March 20, 2017 with `RANK()`.

https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%202/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

FOLIUM FEATURE	COMPONENT TYPE	IMPLEMENTATION & REASON
Circles and popups	CircleMarker	Mark four pads with 1 km circles and show each site name on click.
Offset DivIcon labels	DivIcon	Keep pad names visible where Florida sites sit close.
MarkerCluster	FastMarkerCluster	Group launches that share pad coordinates so markers remain accessible.
Launch markers and popups	Outcome Marker	Show each landing outcome with a green/red icon and identify the launch on click.
MousePosition	Coordinate Plugin	Read coordinates for selected coastline, railway, highway and city points.
Distance labels and PolyLine	PolyLine & Text	Display calculated kilometres and connect each measured point to its launch pad.



https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%203/lab-jupyter-launch-site-location-v2.ipynb

Build a Dashboard with Plotly Dash

[Live dashboard: https://spacex-launch-analytics.netlify.app/](https://spacex-launch-analytics.netlify.app/)

Successful launches by site pie chart

Counts successes at each pad to identify which site has the most.

Selected-site outcome pie chart

Shows success versus failure at one pad so its success rate can be assessed.

Payload and outcome scatter plot

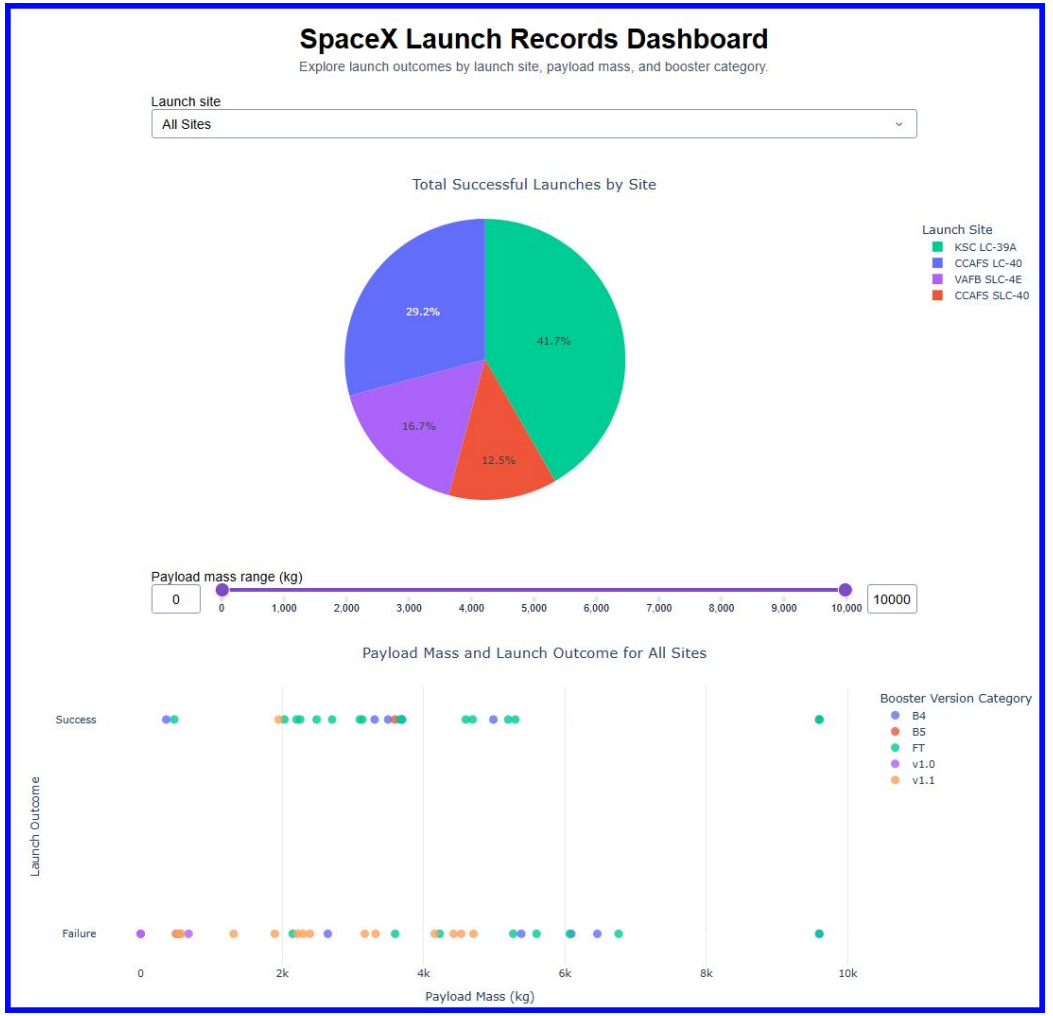
Plots payload mass against success or failure, with color showing booster category, to inspect how outcomes vary across payloads and boosters.

Launch-site dropdown

Switches the pie chart between all sites and one site, and filters the scatter plot for that site.

Payload slider and point hover

Narrows the scatter plot to a mass range; hovering identifies the launch site and booster version behind a point.



https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%203/spacex-dash-app.py

Predictive Analysis (Classification)

TASK 6 | SVM GRID SEARCH

```
best = {
    k: float(v) if hasattr(v, "item") else v
    for k, v in svm_cv.best_params_.items()
}

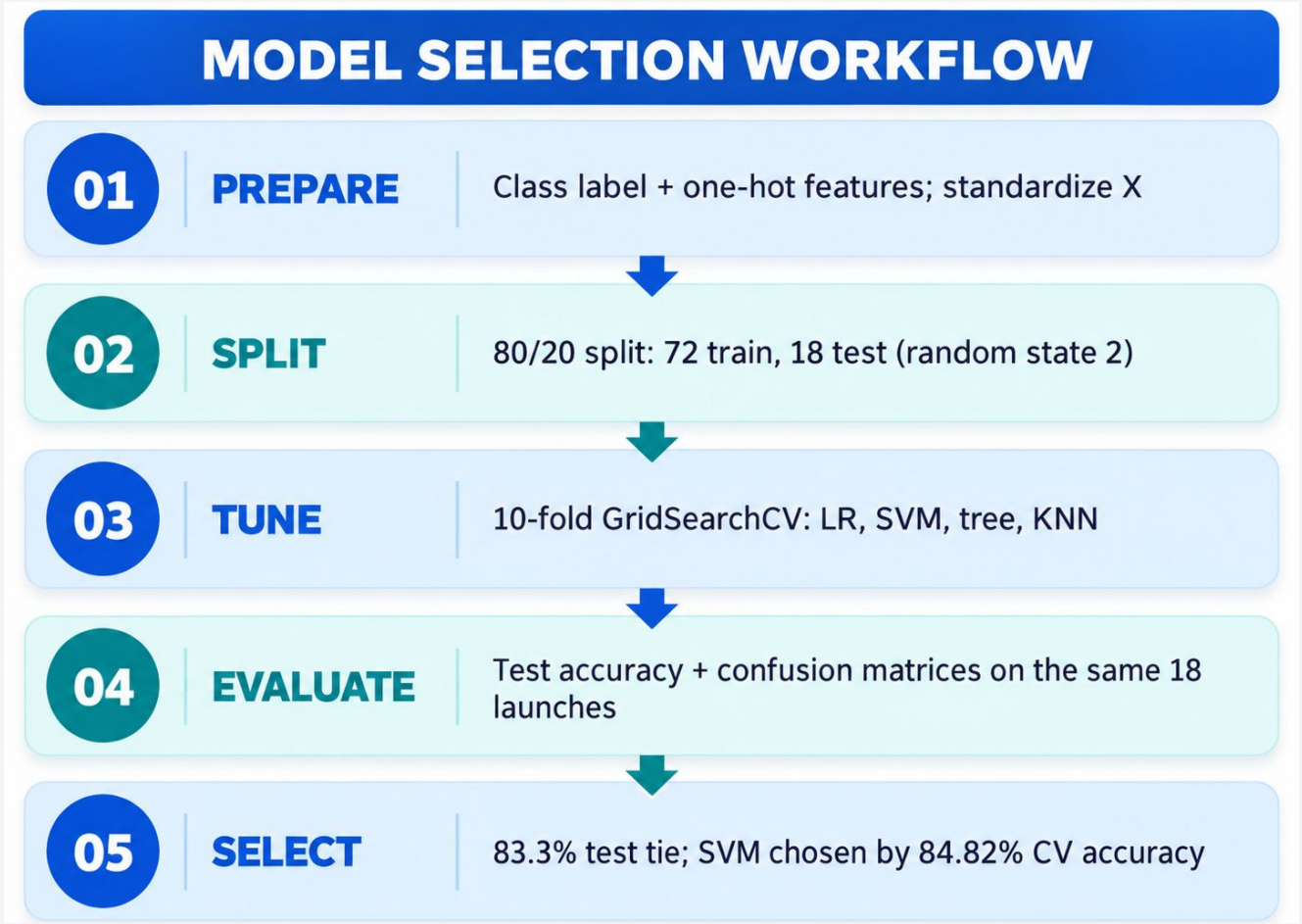
accuracy = svm_cv.best_score_

print("Best Hyperparameters")
print(f"  C       : {best['C']:.2f}")
print(f"  gamma   : {best['gamma']:.5f}")
print(f"  kernel  : {best['kernel']}")
print(f"Best Accuracy : {accuracy:.4f}")
```

✓ 0.0s

```
Best Hyperparameters
  C       : 1.00
  gamma   : 0.03162
  kernel  : sigmoid
Best Accuracy : 0.8482
```

84.82% mean cross-validation accuracy

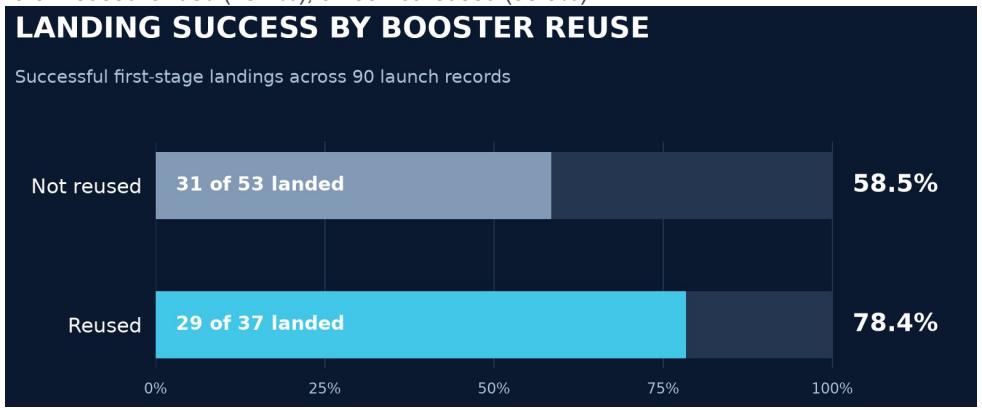


https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%204/SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb

Results

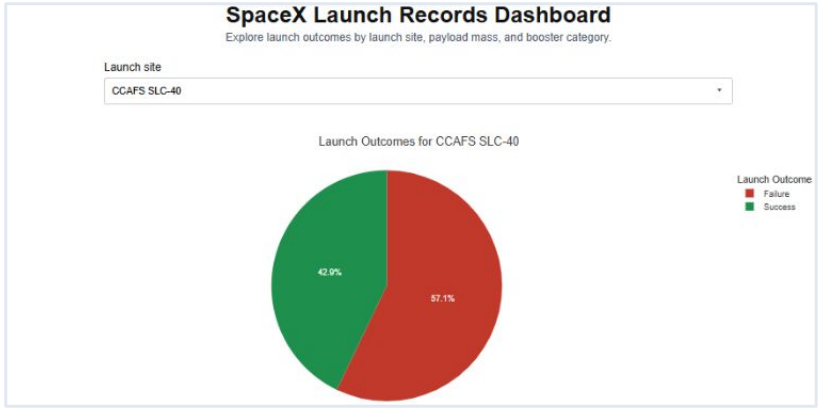
EXPLORATORY ANALYSIS | BOOSTER REUSE

29/37 reused landed (78.4%); 31/53 not reused (58.5%).



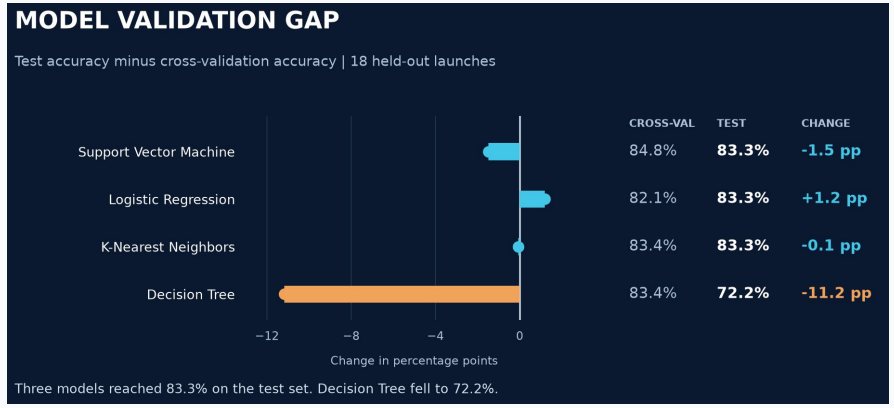
INTERACTIVE DASHBOARD | CCAFS SLC-40

3 of 7 launches had successful landings.



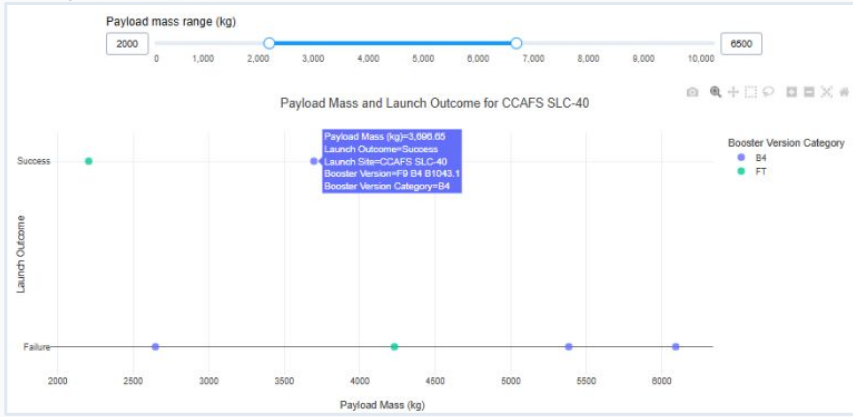
PREDICTIVE ANALYSIS | MODEL VALIDATION

3 models tied at 83.3% test; Decision Tree: 83.4% CV → 72.2% test.



INTERACTIVE DASHBOARD | PAYLOAD & SUCCESS

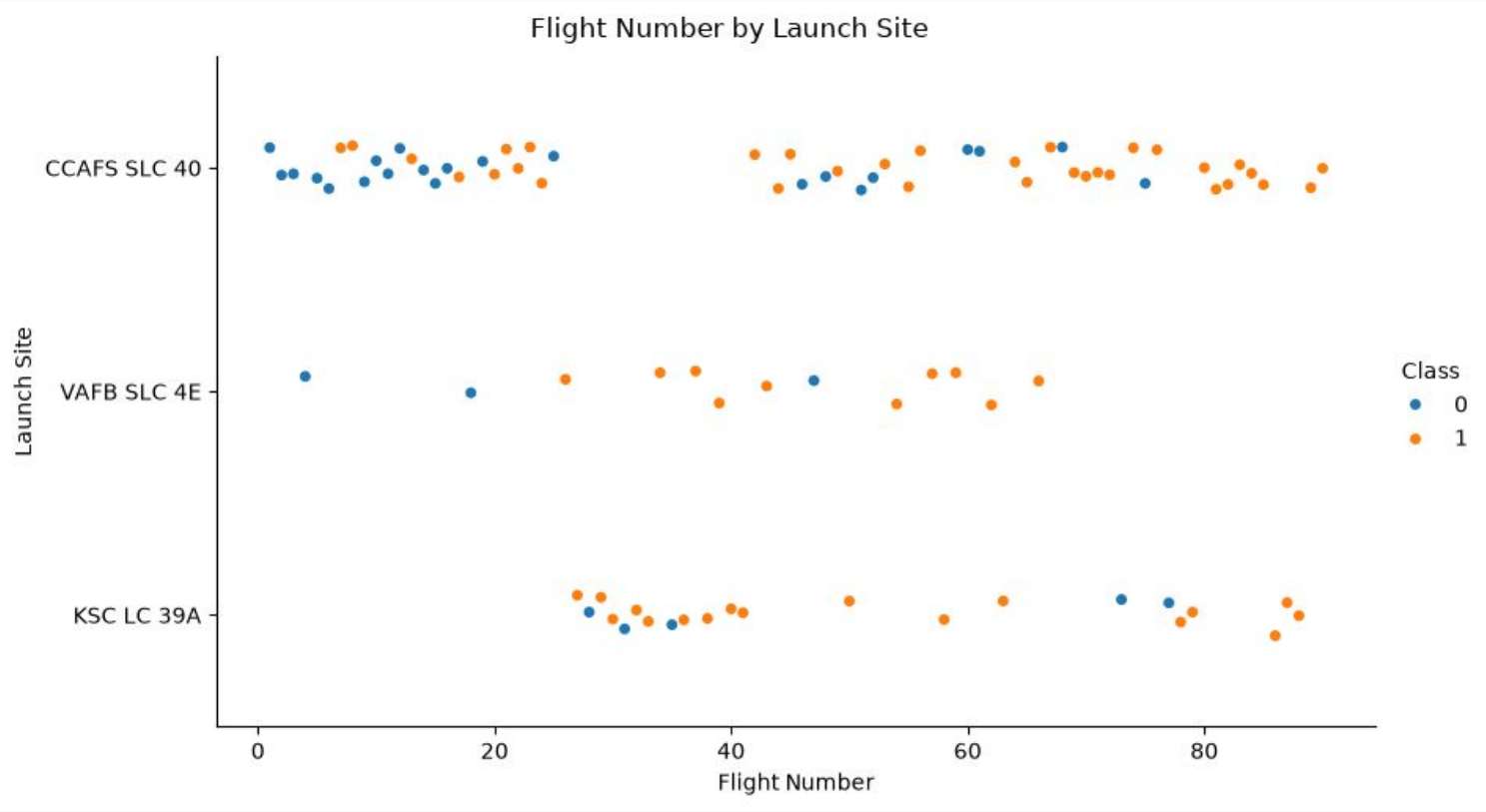
2,000–6,500 kg filter; hover identifies a launch.



[EDA notebook] [Dash Python] [Live dashboard] [ML notebook]

Results | EDA and SQL

EXPLORATORY DATA ANALYSIS



SQL NOTEBOOK MILESTONE

December 22, 2015

First successful ground-pad landing

MISSION OUTCOMES (101-ROW SAMPLE)

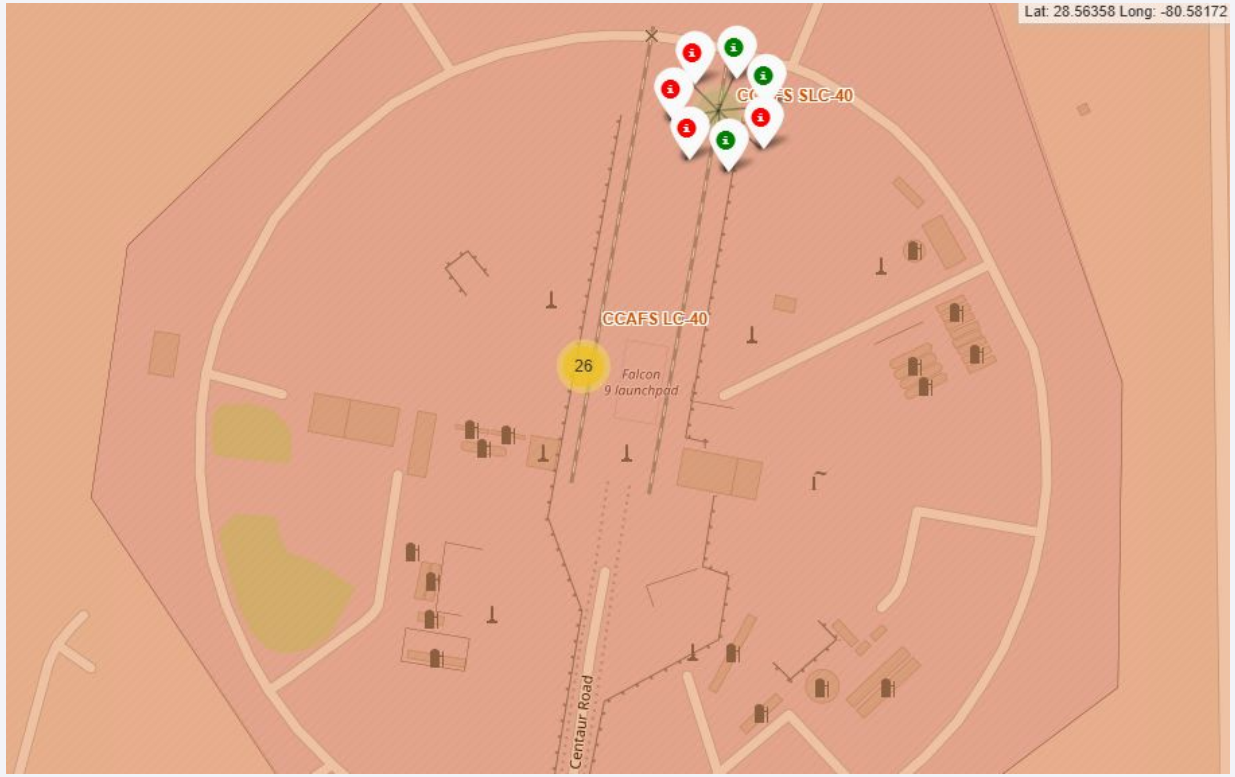
99	**Success** Primary mission objective achieved
1	**Failure** (in flight) Loss during ascent phase
1	**Success** (payload status unclear) Orbit reached, payload unverified

https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%202/jupyter-labs-eda-dataviz-v2.ipynb

https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%202/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Results | Folium Map

LANDING OUTCOMES AT CCAFS LC-40



Green: landing success Red: landing failure

NEARBY FEATURES (KM)

Launch Site	Proximity	LatLaunchSite	LongLaunchSite	Lat	Long	distance
CCAFS LC-40	coastline	28.562302	-80.577356	28.56340	-80.56796	0.926054
	railway	28.562302	-80.577356	28.57217	-80.58528	1.343093
	highway	28.562302	-80.577356	28.56301	-80.57076	0.649222
	city	28.562302	-80.577356	28.10469	-80.64784	51.365738
CCAFS SLC-40	coastline	28.563197	-80.576820	28.56409	-80.56806	0.861525
	railway	28.563197	-80.576820	28.57217	-80.58528	1.295798
	highway	28.563197	-80.576820	28.56385	-80.57088	0.584818
	city	28.563197	-80.576820	28.10469	-80.64784	51.471476

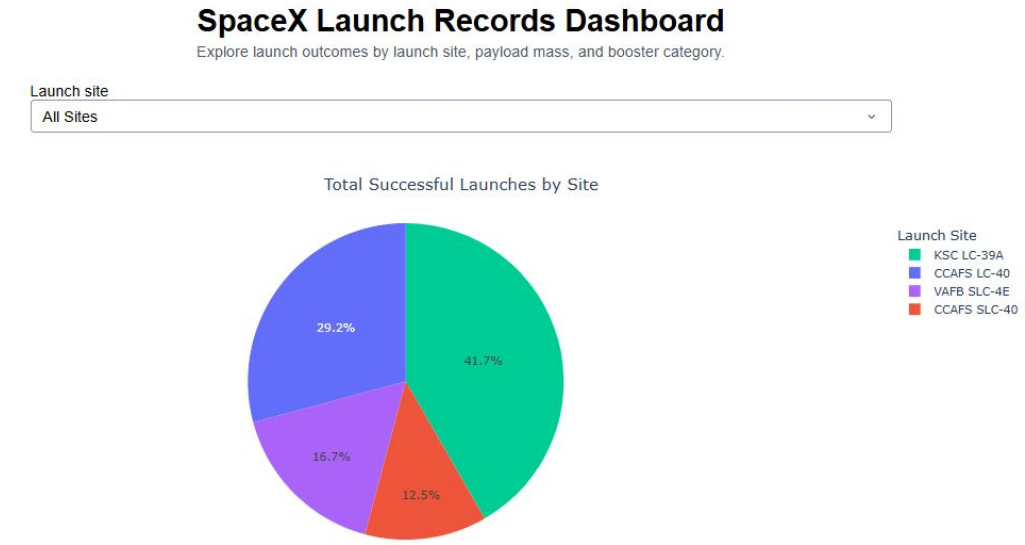
The selected coast and highway points are less than 1 km from both CCAFS sites.

https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%203/lab-jupyter-launch-site-location-v2.ipynb

Results | Interactive Analytics

Changing the site filter changes the landing outcome view.

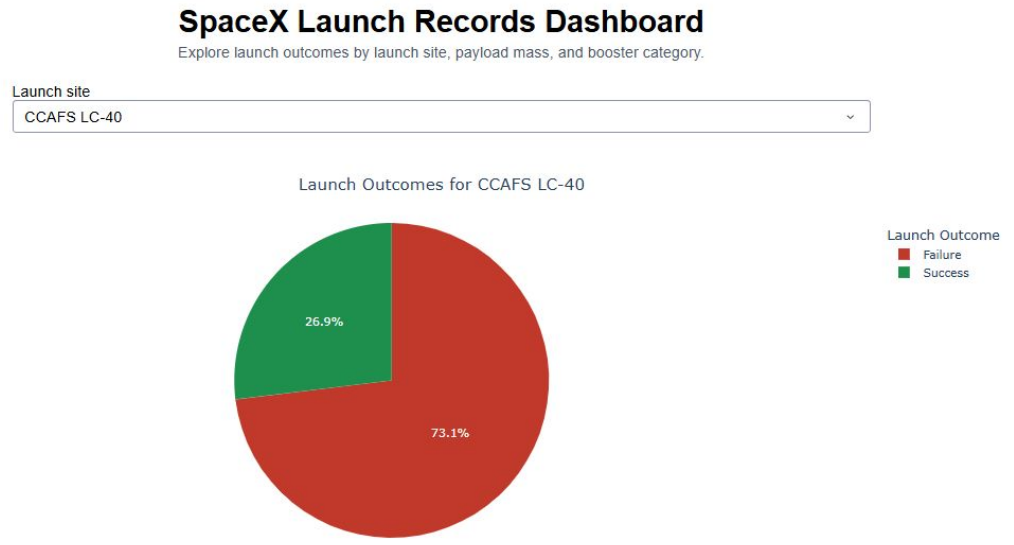
ALL SITES



41.7%

KSC LC-39A accounts for 41.7% of successful landings.

CCAFS LC-40



26.9%

At CCAFS LC-40, 26.9% of launches landed successfully.

https://spacex-launch-analytics.netlify.app/

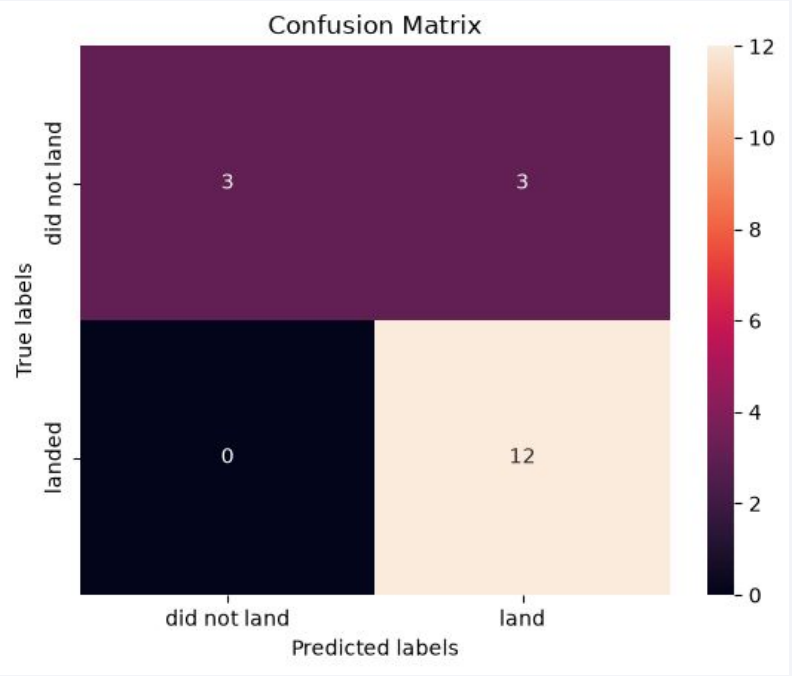
01-all-sites-dashboard.png

02-selected-site-dashboard.png

Results | Predictive Models

Three models reached 83.3% test accuracy.

MODEL	ACCURACY
Logistic Regression	83.3%
Support Vector Machine	83.3%
K-Nearest Neighbors	83.3%
Decision Tree	72.2%

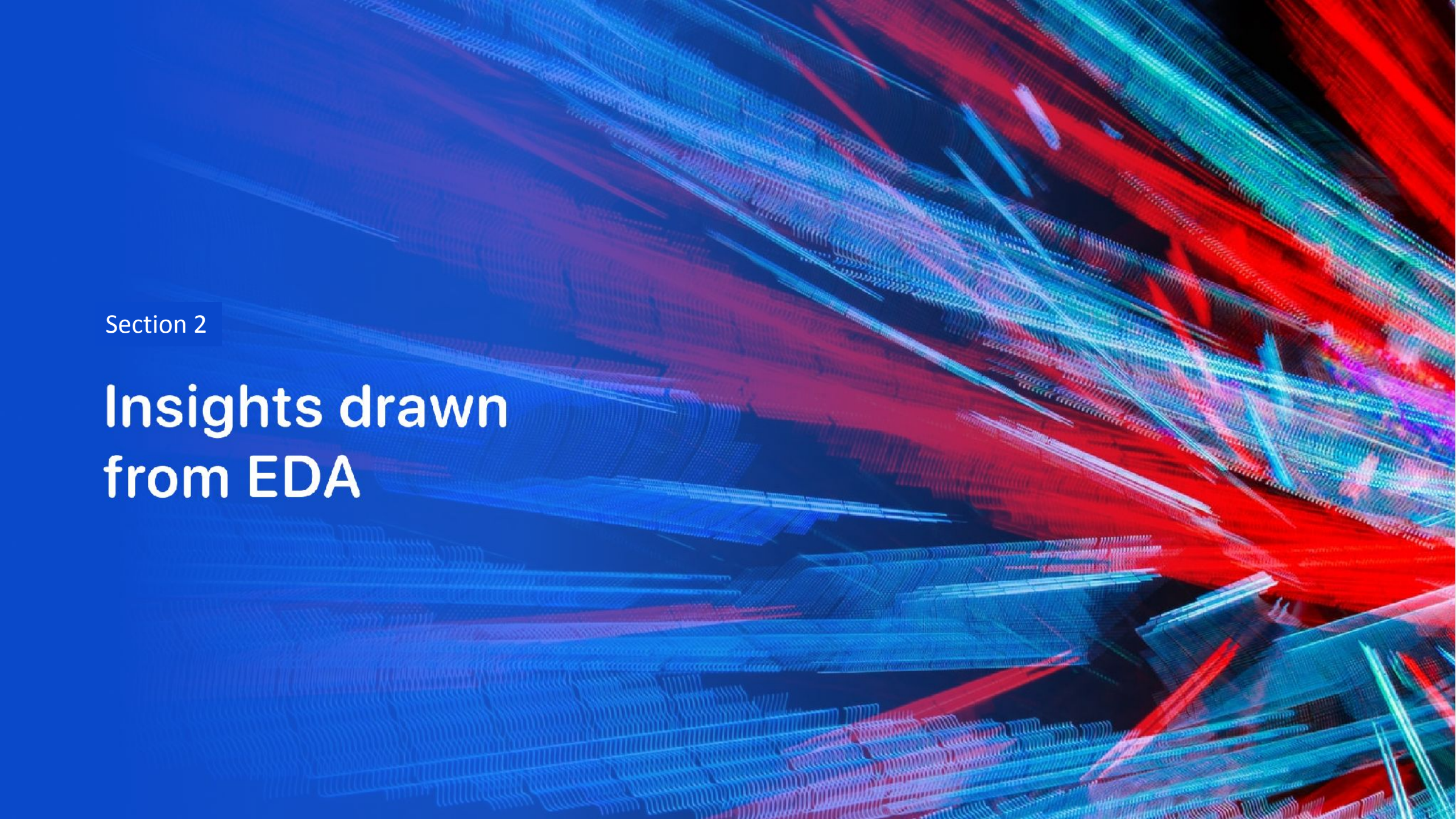


The selected SVM found all 12 successes, but missed 3 of 6 failures.

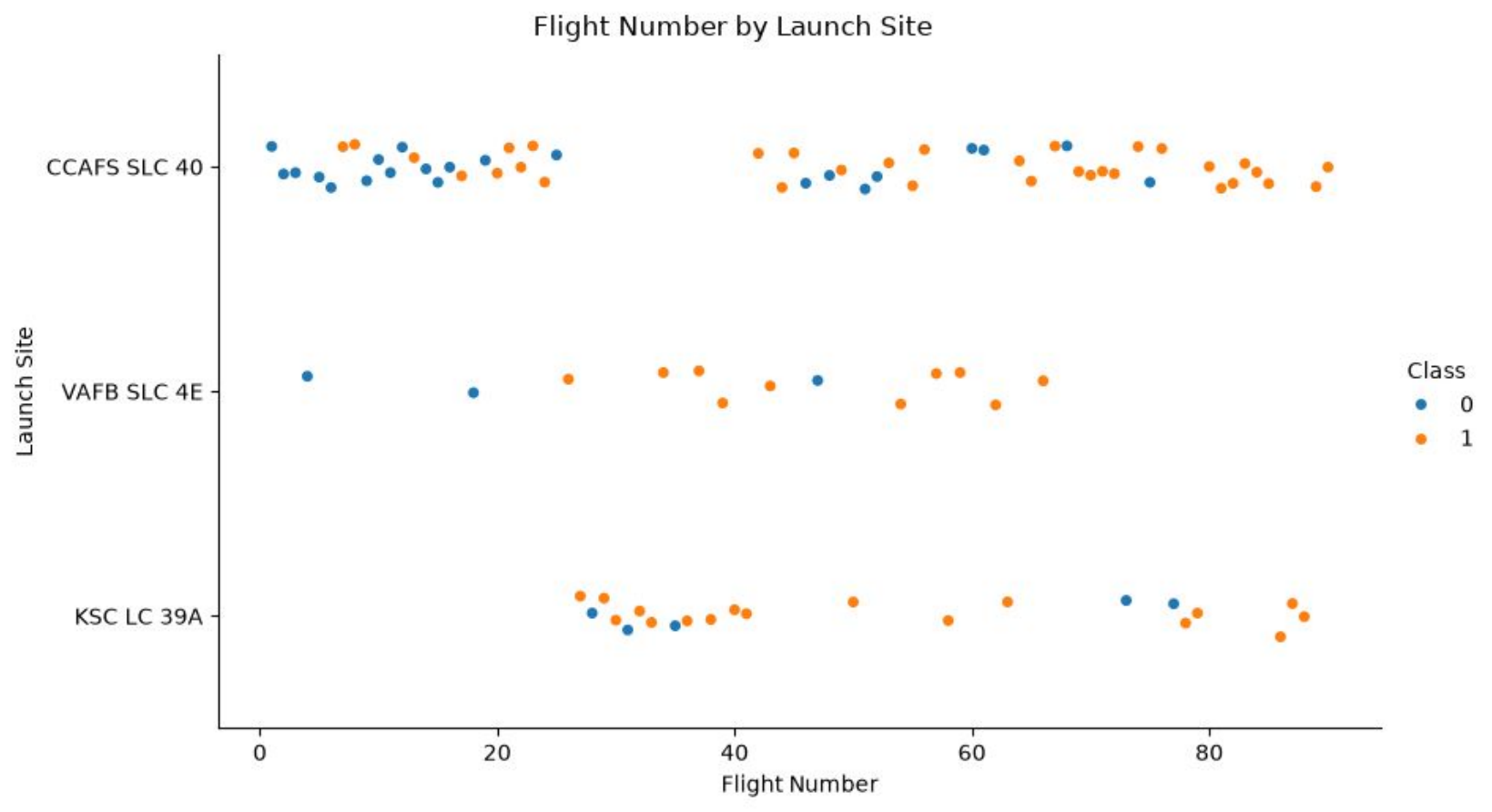
https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%204/SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb

Section 2

Insights drawn from EDA



Flight Number vs. Launch Site



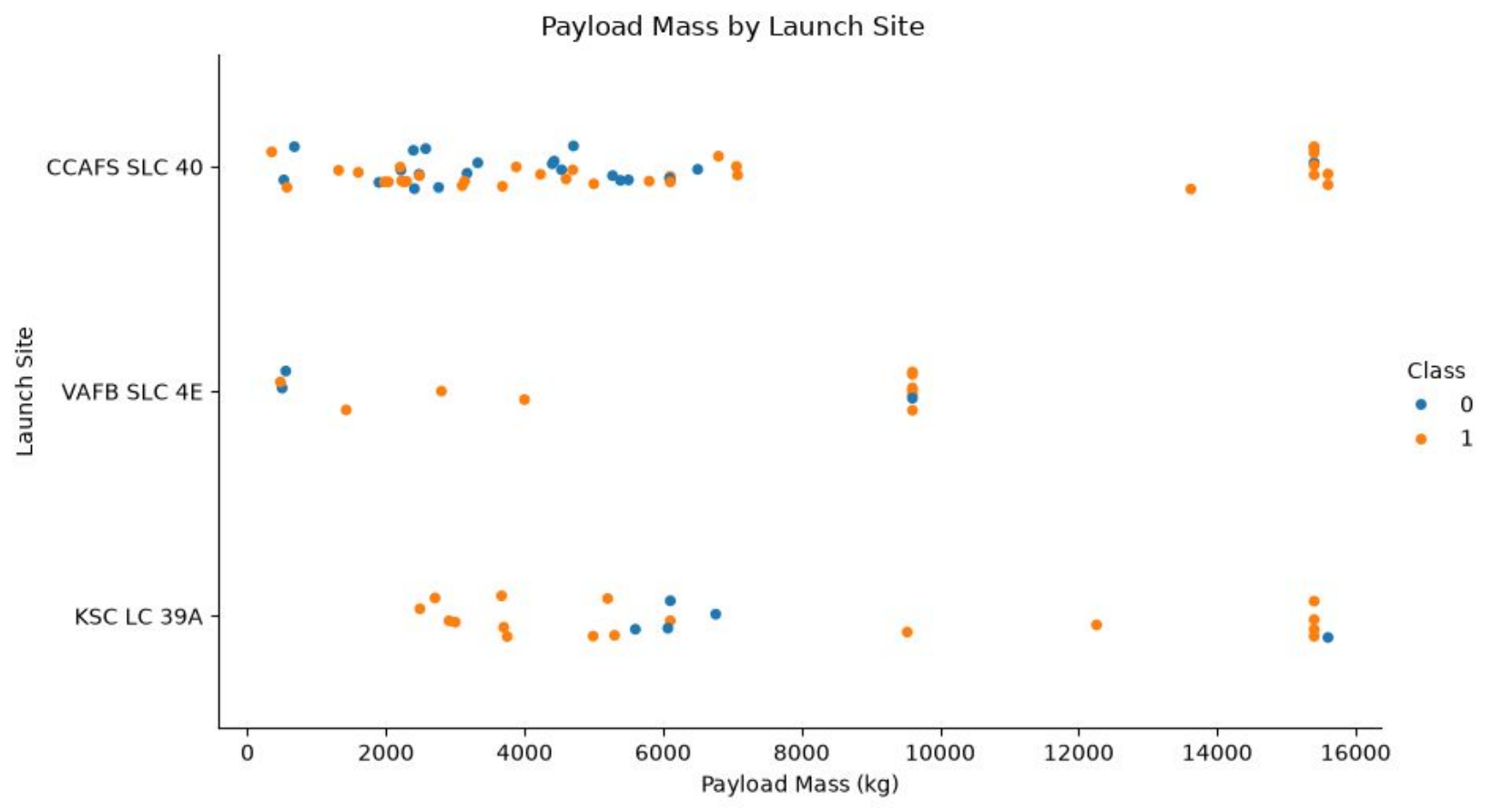
LATER FLIGHTS

Successful landings appear more often as flight number rises.

Orange (Class 1) marks a successful landing.

https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%202/jupyter-labs-eda-dataviz-v2.ipynb

Payload Mass vs. Launch Site



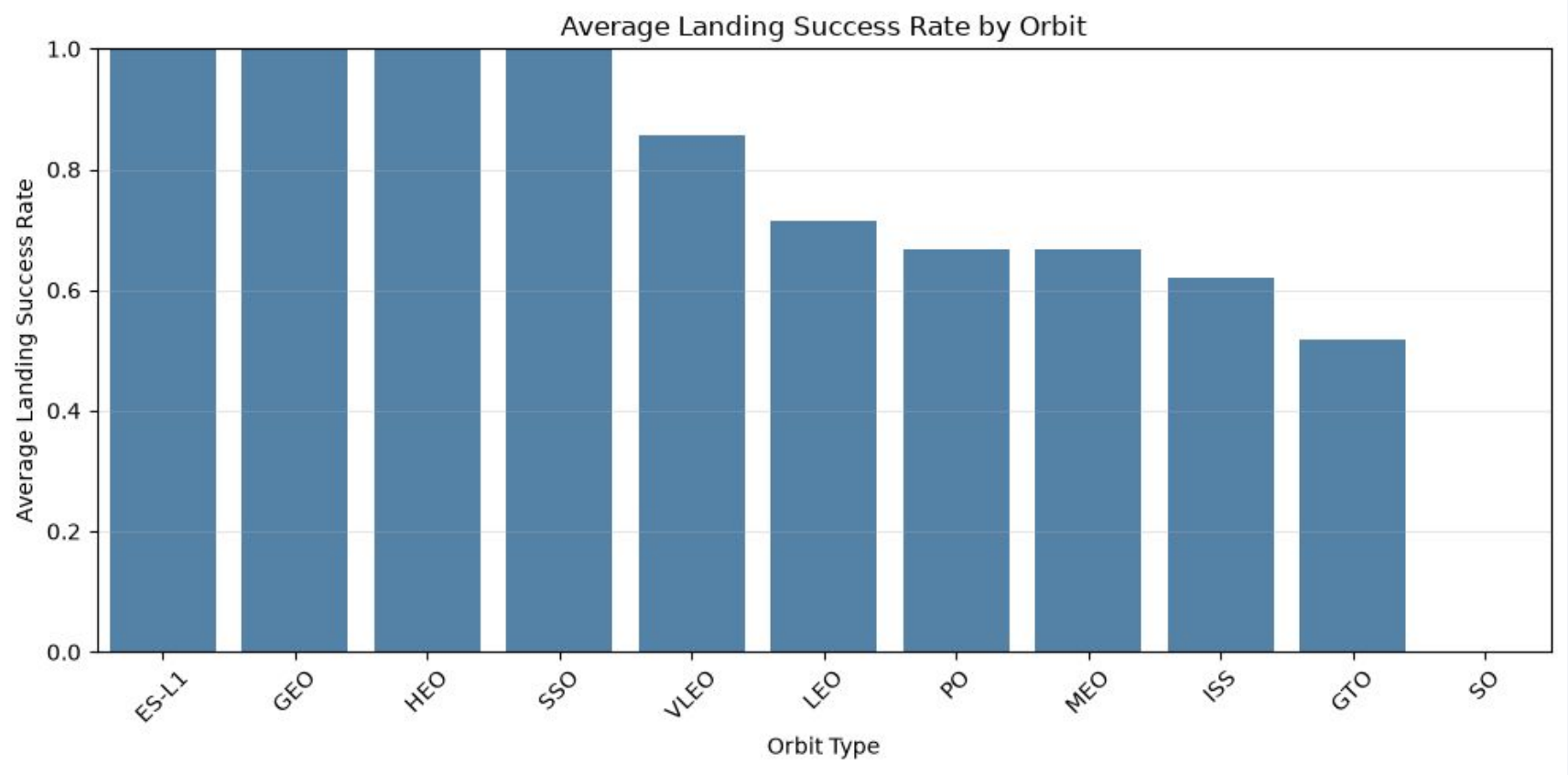
9,600 KG

The heaviest VAFB payload in this sample is 9,600 kg.

Payload mass alone does not separate landing outcomes.

https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%202/jupyter-labs-eda-dataviz-v2.ipynb

Landing Success Rate by Orbit



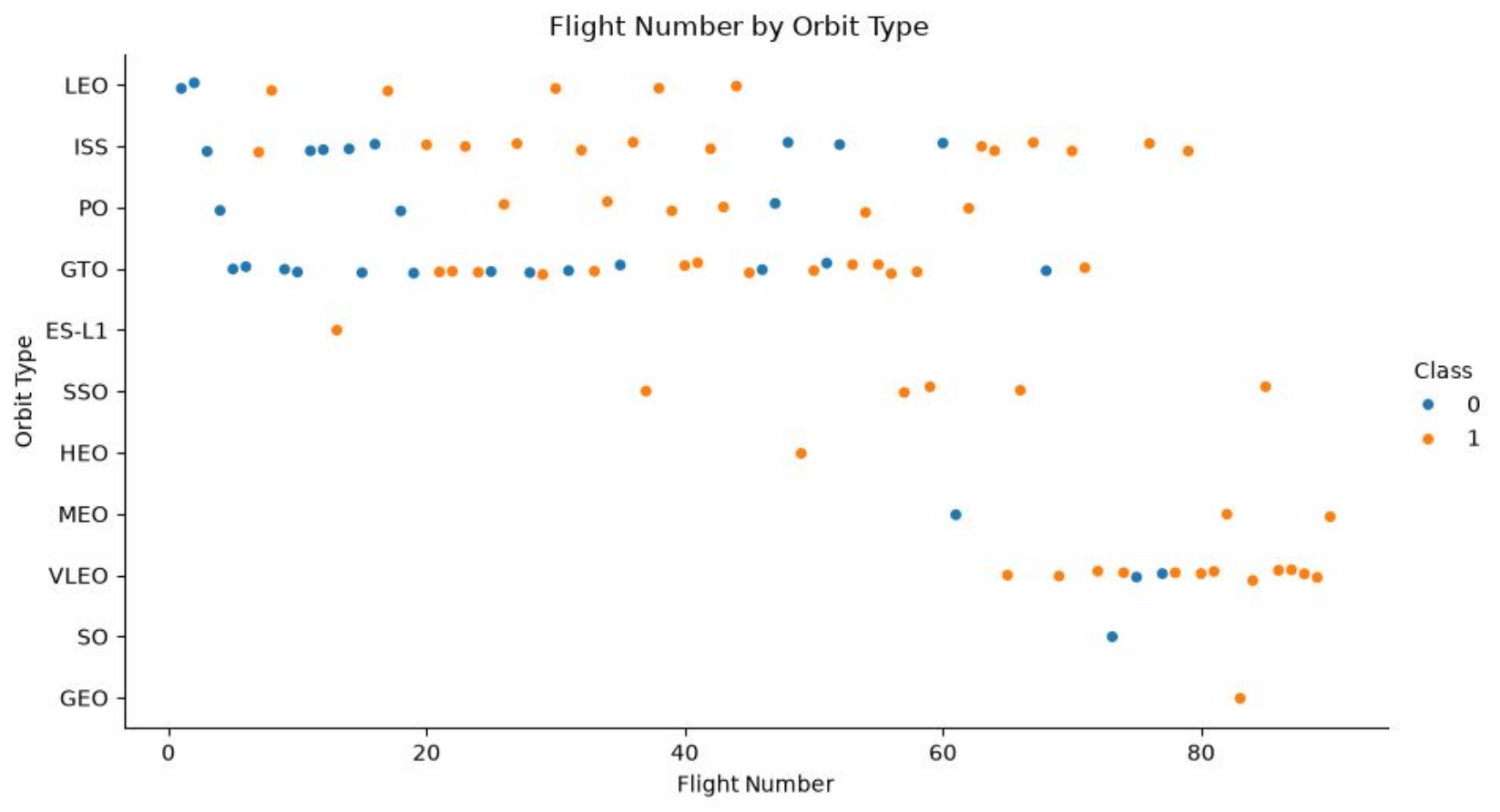
86% VS. 52%

VLEO: 12 of 14 landed.
GTO: 14 of 27 landed.

Three 100% orbit bars each represent one flight.

https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%202/jupyter-labs-eda-dataviz-v2.ipynb

Flight Number vs. Orbit Type



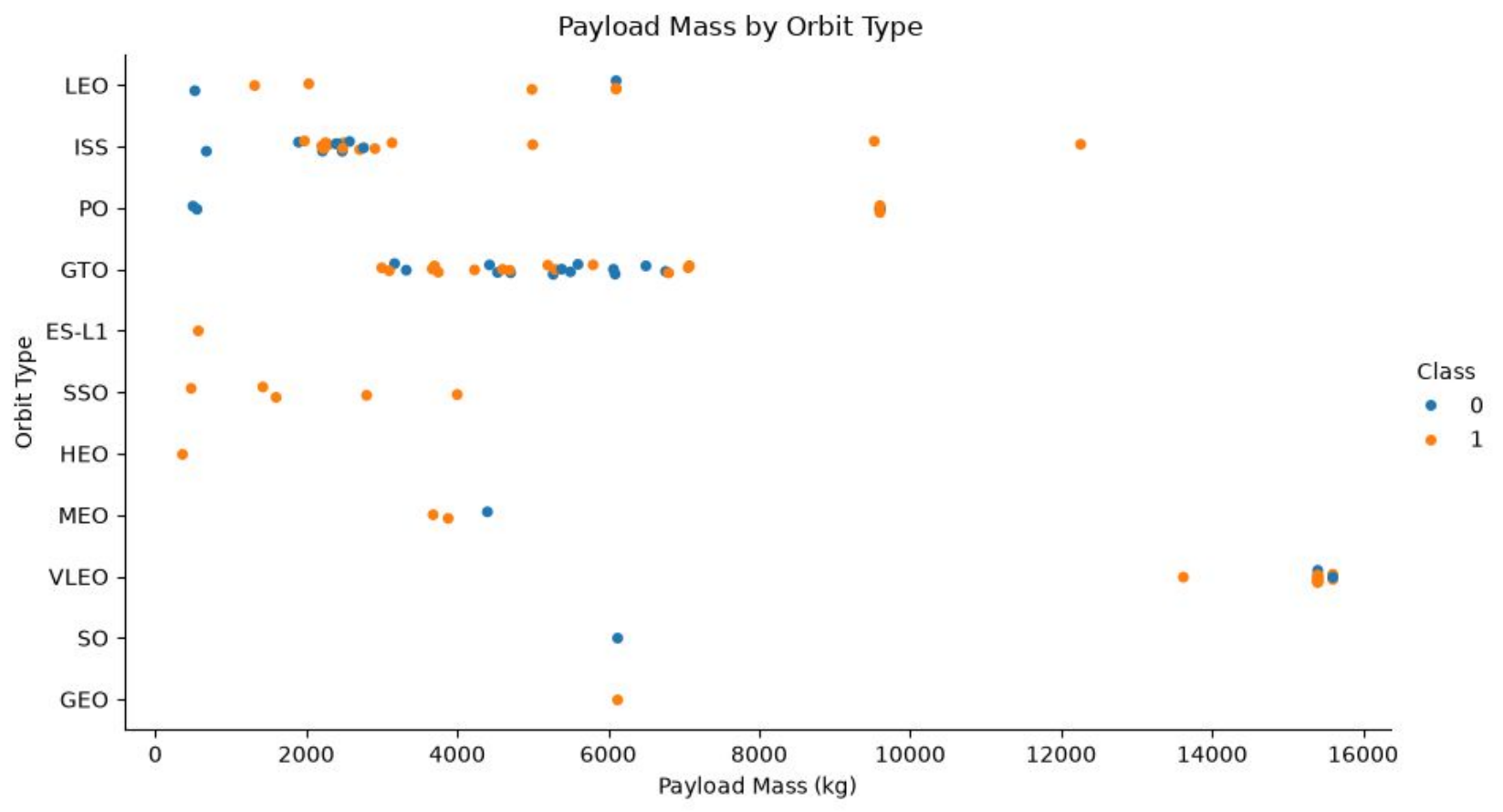
LEO / GTO

Later LEO flights tend to succeed. GTO stays mixed across flight numbers.

Orange (Class 1) marks a successful landing.

https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%202/jupyter-labs-eda-dataviz-v2.ipynb

Payload Mass vs. Orbit Type



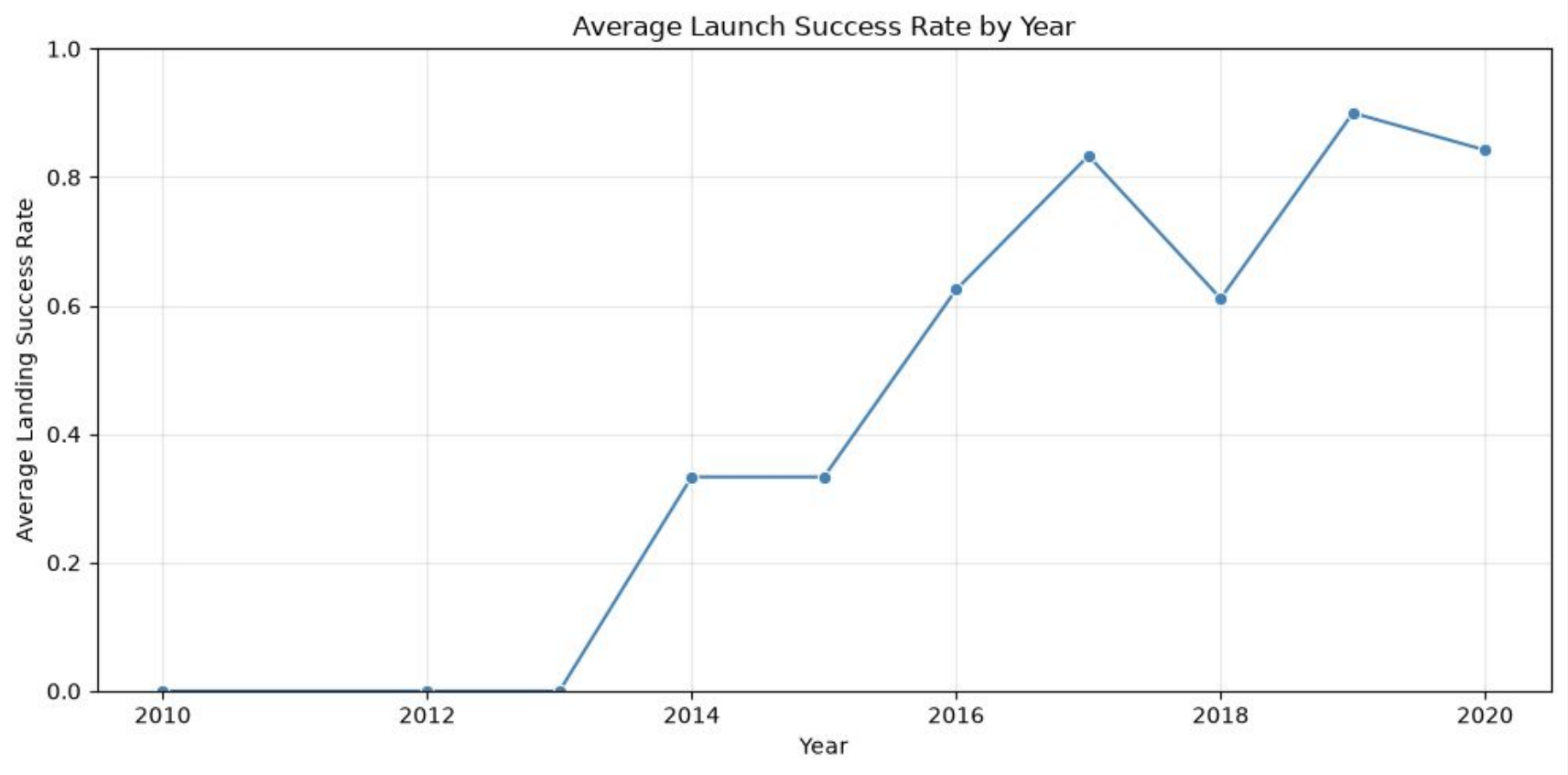
GTO

Both landing outcomes appear across its payload range.

Payload mass does not show a clear success cutoff.

https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%202/jupyter-labs-eda-dataviz-v2.ipynb

Landing Success Rate by Year



90% IN 2019

After the 2018 dip, landing success passed 80% in both 2019 and 2020.

2019: 9 of 10
2020: 16 of 19

https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%202/jupyter-labs-eda-dataviz-v2.ipynb

All Launch Site Names

SQL QUERY RESULT

01 CCAFS LC-40

02 CCAFS SLC-40

03 KSC LC-39A

04 VAFB SLC-4E

Task 1

Display the names of the unique launch sites in the space mission

```
%%sql
WITH normalized_launches AS (
    SELECT NULLIF(TRIM("Launch_Site"), '') AS launch_site
    FROM SPACEXTABLE
)
SELECT DISTINCT launch_site
FROM normalized_launches
WHERE launch_site IS NOT NULL
ORDER BY launch_site;
```

```
 * sqlite:///my_data1.db
Done.
```

launch_site
CCAFS LC-40
CCAFS SLC-40
KSC LC-39A
VAFB SLC-4E

https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%202/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Five Launch Records at CCAFS

5 records where launch_site begins with CCA

06-04-2010	F9 v1.0 B0003	CCAFS LC-40	Failure (parachute)
12-08-2010	F9 v1.0 B0004	CCAFS LC-40	Failure (parachute)
05-22-2012	F9 v1.0 B0005	CCAFS LC-40	No attempt
10-08-2012	F9 v1.0 B0006	CCAFS LC-40	No attempt
03-01-2013	F9 v1.0 B0007	CCAFS LC-40	No attempt

5

Early CCAFS Missions

All five missions succeeded. None recorded a successful landing.

Task 2 SQL output

launch_date	launch_time_utc	booster_version	launch_site	payload	payload_mass_kg	orbit	customer	mission_outcome	landing_outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0.0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0.0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525.0	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500.0	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677.0	LEO (ISS)	NASA (CRS)	Success	No attempt

https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%202/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Total Payload Mass for NASA (CRS)

TOTAL PAYLOAD MASS

45,596 kg

NASA (CRS) missions carried this total payload mass in the SQL result.

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

```
%%sql
WITH normalized_launches AS (
    SELECT
        TRIM(Customer) AS customer,
        CAST(PAYLOAD_MASS__KG_ AS REAL) AS payload_mass_kg
    FROM SPACEXTABLE
)
SELECT ROUND(SUM(payload_mass_kg), 2) AS total_payload_mass_kg
FROM normalized_launches
WHERE customer = 'NASA (CRS)';
```

```
 * sqlite:///my_data1.db
Done.
```

total_payload_mass_kg
45596.0

https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%202/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Average Payload Mass for F9 v1.1

AVERAGE PAYLOAD MASS

2,928.4 kg

F9 v1.1 launches carried this average payload mass in the SQL result.

Task 4

Display average payload mass carried by booster version F9 v1.1

```
%%sql
WITH normalized_launches AS (
    SELECT
        TRIM("Booster_Version") AS booster_version,
        CAST(PAYLOAD_MASS__KG_ AS REAL) AS payload_mass_kg
    FROM SPACEXTABLE
)
SELECT ROUND(AVG(payload_mass_kg), 2) AS average_payload_mass_kg
FROM normalized_launches
WHERE booster_version = 'F9 v1.1';
```

```
 * sqlite:///my_data1.db
Done.
```

average_payload_mass_kg
2928.4

https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%202/jupyter-labs-eda-sql-coursera_sqllite.ipynb

First Successful Ground-Pad Landing

FIRST SUCCESSFUL GROUND-PAD LANDING

December 22, 2015

This is the earliest launch date with a successful ground-pad landing outcome.

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

```
%%sql
WITH normalized_launches AS (
    SELECT
        Date AS launch_date,
        TRIM("Landing_Outcome") AS landing_outcome
    FROM SPACEXTABLE
)
SELECT MIN(launch_date) AS first_successful_ground_pad_landing_date
FROM normalized_launches
WHERE landing_outcome = 'Success (ground pad)';
```

```
 * sqlite:///my_data1.db
Done.
```

first_successful_ground_pad_landing_date
2015-12-22

https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%202/jupyter-labs-eda-sql-coursera_sqllite.ipynb

SUCCESSFUL DRONE-SHIP LANDINGS

Drone-Ship Landings: 4,000–6,000 kg

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
%%sql
WITH normalized_launches AS (
    SELECT
        TRIM("Booster_Version") AS booster_version,
        TRIM("Landing_Outcome") AS landing_outcome,
        CAST(PAYLOAD_MASS__KG_ AS REAL) AS payload_mass_kg
    FROM SPACEXTABLE
)
SELECT DISTINCT booster_version
FROM normalized_launches
WHERE landing_outcome = 'Success (drone ship)'
  AND payload_mass_kg > 4000
  AND payload_mass_kg < 6000
ORDER BY booster_version;
```

```
 * sqlite:///my_data1.db
Done.
```

booster_version
F9 FT B1021.2
F9 FT B1031.2
F9 FT B1022
F9 FT B1026

01
F9 FT B1021.2
Falcon 9 Full Thrust • Drone Ship Landing

02
F9 FT B1031.2
Falcon 9 Full Thrust • Drone Ship Landing

03
F9 FT B1022
Falcon 9 Full Thrust • Drone Ship Landing

04
F9 FT B1026
Falcon 9 Full Thrust • Drone Ship Landing

https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%202/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Mission Outcomes in the SQL Dataset

MISSION OUTCOME	COUNT
Success	99
Failure (in flight)	1
Success (payload status unclear)	1

The SQL result counts mission outcomes across 101 launch records.

Task 7

List the total number of successful and failure mission outcomes

```
%%sql
WITH normalized_launches AS (
    SELECT NULLIF(TRIM("Mission_Outcome"), '') AS mission_outcome
    FROM SPACEXTABLE
)
SELECT
    mission_outcome,
    COUNT(*) AS outcome_count
FROM normalized_launches
WHERE mission_outcome IS NOT NULL
GROUP BY mission_outcome
ORDER BY outcome_count DESC, mission_outcome;
```

```
 * sqlite:///my_data1.db
Done.
```

mission_outcome	outcome_count
Success	99
Failure (in flight)	1
Success (payload status unclear)	1

https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%202/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Boosters with the Maximum Payload

12 BOOSTER-VERSION ENTRIES IN THE SQL RESULT

01	F9 B5 B1048.4	07	F9 B5 B1051.4
02	F9 B5 B1048.5	08	F9 B5 B1051.6
03	F9 B5 B1049.4	09	F9 B5 B1056.4
04	F9 B5 B1049.5	10	F9 B5 B1058.3
05	F9 B5 B1049.7	11	F9 B5 B1060.2
06	F9 B5 B1051.3	12	F9 B5 B1060.3

Task 8

List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

```
%%sql
WITH normalized_launches AS (
    SELECT
        TRIM("Booster_Version") AS booster_version,
        CAST(PAYLOAD_MASS__KG_ AS REAL) AS payload_mass_kg
    FROM SPACEXTABLE
),
max_payload AS (
    SELECT MAX(payload_mass_kg) AS maximum_payload_mass_kg
    FROM normalized_launches
)
SELECT DISTINCT normalized_launches.booster_version
FROM normalized_launches
JOIN max_payload
  ON normalized_launches.payload_mass_kg = max_payload.maximum_payload_mass_kg
ORDER BY normalized_launches.booster_version;
```

```
 * sqlite:///my_data1.db
Done.
```

booster_version
F9 B5 B1048.4
F9 B5 B1048.5
F9 B5 B1049.4
F9 B5 B1049.5
F9 B5 B1049.7
F9 B5 B1051.3
F9 B5 B1051.4
F9 B5 B1051.6
F9 B5 B1056.4
F9 B5 B1058.3
F9 B5 B1060.2
F9 B5 B1060.3

All 12 results are F9 B5 booster-version entries.

https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%202/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Failed Drone-Ship Landings in 2015

TWO MATCHING SQL RECORDS

JANUARY 2015	APRIL 2015
F9 v1.1 B1012	F9 v1.1 B1015
CCAFS LC-40	CCAFS LC-40
Failure (drone ship)	Failure (drone ship)

Task 9

List the records which will display the month names, failure landing outcomes in drone ship, booster versions, and launch site for the months in year 2015.

```
%%sql
WITH normalized_launches AS (
    SELECT
        Date AS launch_date,
        TRIM("Landing_Outcome") AS landing_outcome,
        TRIM("Booster_Version") AS booster_version,
        TRIM("Launch_Site") AS launch_site
    FROM SPACEXTABLE
)
SELECT
    substr(launch_date, 6, 2) AS month,
    landing_outcome,
    booster_version,
    launch_site
FROM normalized_launches
WHERE launch_date >= '2015-01-01'
  AND launch_date < '2016-01-01'
  AND landing_outcome = 'Failure (drone ship)'
ORDER BY launch_date;
```

```
 * sqlite:///my_data1.db
Done.
```

month	landing_outcome	booster_version	launch_site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Both 2015 drone-ship failures in the result launched from CCAFS LC-40.

https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%202/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Landing Outcomes Ranked by Count

No attempt is the most common landing outcome, with 10 records.

2010-06-04 TO 2017-03-20

LANDING OUTCOME	COUNT	RANK
No attempt	10	1
Failure (drone ship)	5	2
Success (drone ship)	5	2
Controlled (ocean)	3	4
Success (ground pad)	3	4
Failure (parachute)	2	6
Uncontrolled (ocean)	2	6
Precluded (drone ship)	1	8

Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the dates 2010-06-04 and 2017-03-20, in descending order.

```
%%sql
WITH normalized_launches AS (
    SELECT
        Date AS launch_date,
        NULLIF(TRIM("Landing_Outcome"), '') AS landing_outcome
    FROM SPACEXTABLE
)
SELECT
    landing_outcome,
    COUNT(*) AS outcome_count,
    RANK() OVER (ORDER BY COUNT(*) DESC) AS outcome_rank
FROM normalized_launches
WHERE launch_date BETWEEN '2010-06-04' AND '2017-03-20'
  AND landing_outcome IS NOT NULL
GROUP BY landing_outcome
ORDER BY outcome_rank, landing_outcome;
```

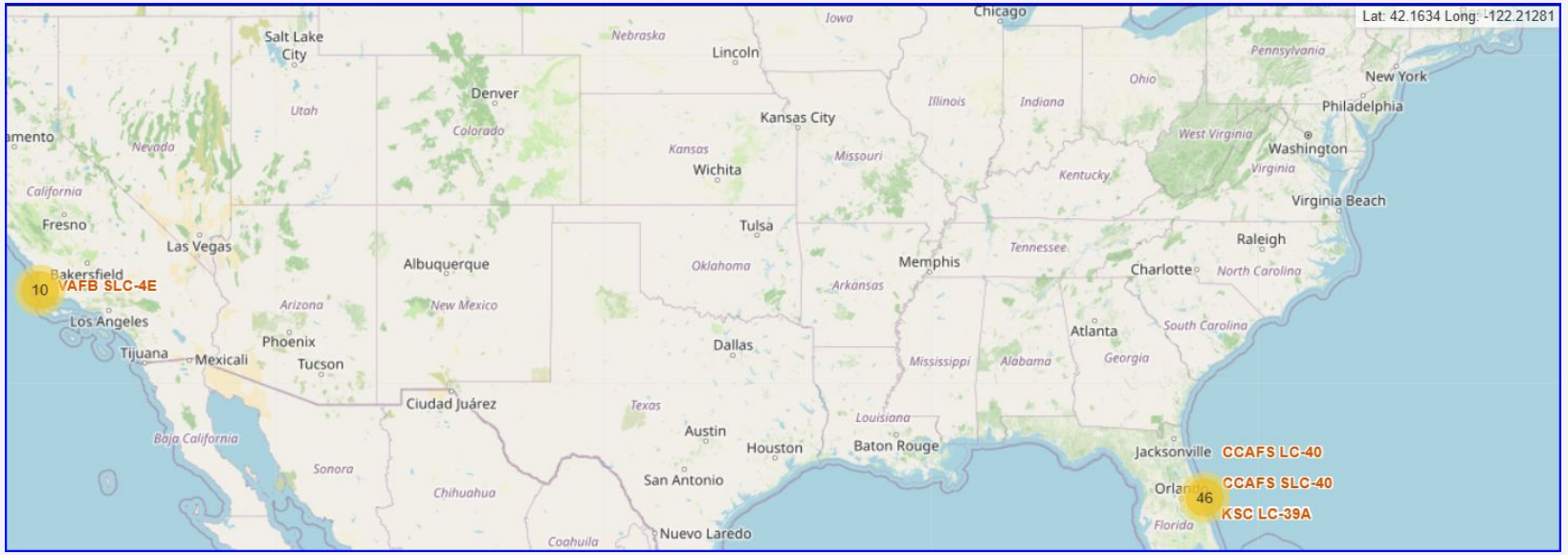
```
 * sqlite:///my_data1.db
Done.
```

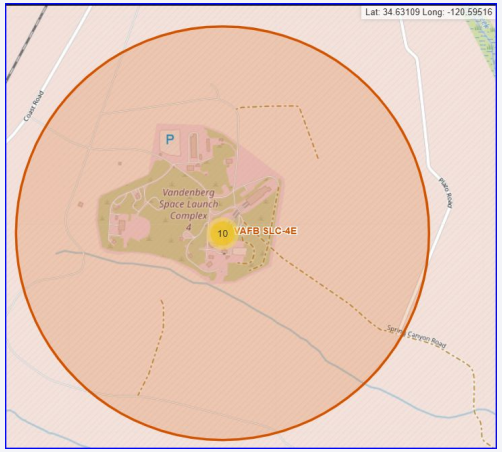
landing_outcome	outcome_count	outcome_rank
No attempt	10	1
Failure (drone ship)	5	2
Success (drone ship)	5	2
Controlled (ocean)	3	4
Success (ground pad)	3	4
Failure (parachute)	2	6
Uncontrolled (ocean)	2	6
Precluded (drone ship)	1	8

https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%202/jupyter-labs-eda-sql-coursera_sqllite.ipynb


Section 3
Launch Sites
Proximities Analysis

Launch Sites and Record Clusters







LOCATIONS

VAFB SLC-4E is in California; KSC LC-39A and the two CCAFS pads are in Florida.

LAUNCH RECORDS

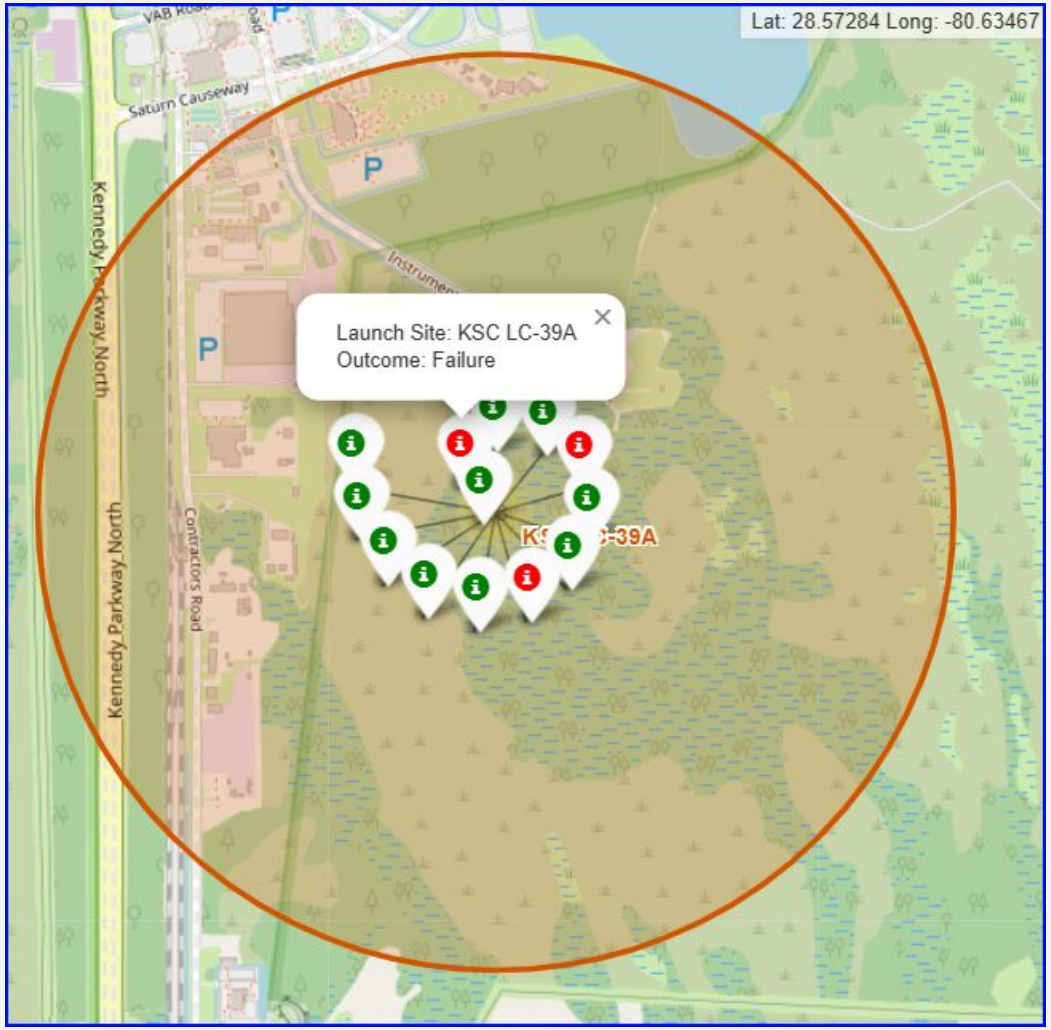
Yellow circles group launch records: 10 at VAFB and 46 in Florida. The Florida close-up splits those 46 into 13 at KSC and 33 at the two CCAFS pads.

MAP KEY

Orange rings show a 1 km radius around each site. The upper-right coordinates show the cursor position when each map was captured.

https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%203/lab-jupyter-launch-site-location-v2.ipynb

Landing Outcomes at KSC LC-39A



LAUNCH SITE METRICS

KSC LC-39A Launch Complex

13 mapped launch records analyzed in this dataset

10
Successful Landings
76.9% site success rate

3
Failed Landings
23.1% landing failure rate

READING THE MAP

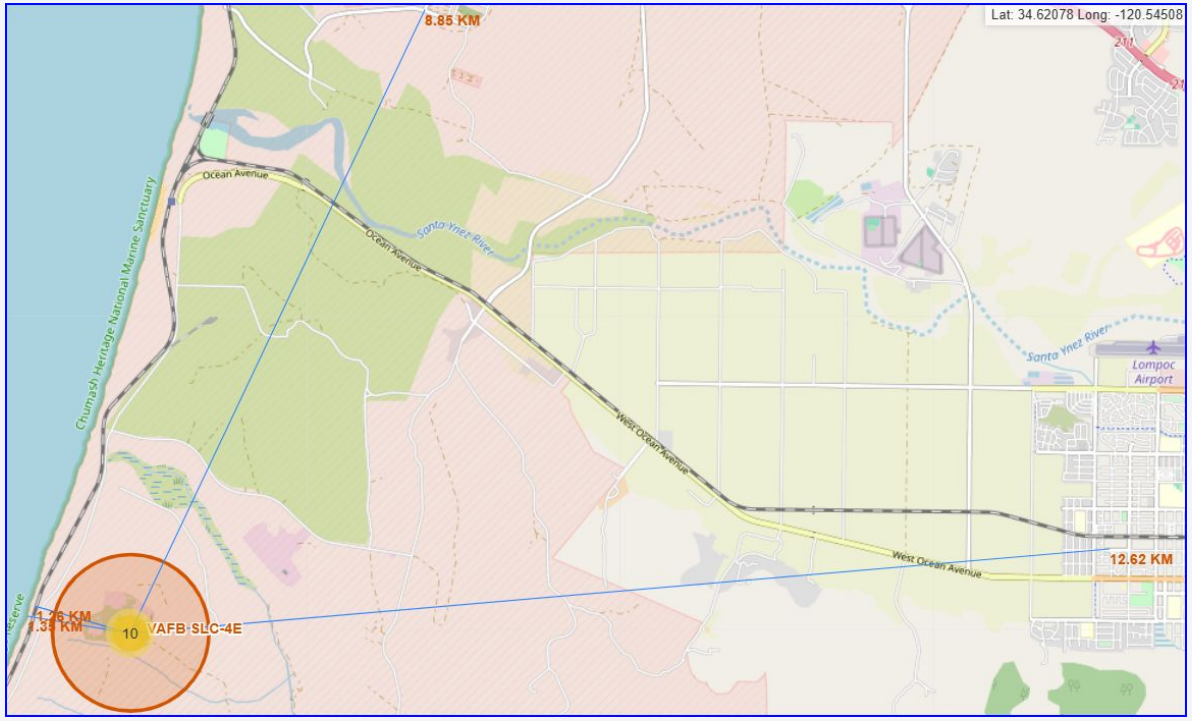
Each marker pin represents an individual launch event. **Green** indicates a successful first-stage landing, while **red** denotes a landing failure. The expanded popup details a recorded failure.

GEOGRAPHIC DETAILS

The orange boundary ring illustrates a 1 km proximity perimeter around the launch complex. Upper-right coordinates mark the active cursor position in this capture.

https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%203/lab-jupyter-launch-site-location-v2.ipynb

VAFB SLC-4E: Proximity to Key Features



Launch Site	Proximity	LatLaunchSite	LongLaunchSite	Lat	Long	distance
CCAFS LC-40	coastline	28.562302	-80.577356	28.56340	-80.56796	0.926054
	railway	28.562302	-80.577356	28.57217	-80.58528	1.343093
	highway	28.562302	-80.577356	28.56301	-80.57076	0.649222
	city	28.562302	-80.577356	28.10469	-80.64784	51.365738
CCAFS SLC-40	coastline	28.563197	-80.576820	28.56409	-80.56806	0.861525
	railway	28.563197	-80.576820	28.57217	-80.58528	1.295798
	highway	28.563197	-80.576820	28.56385	-80.57088	0.584818
	city	28.563197	-80.576820	28.10469	-80.64784	51.471476
KSC LC-39A	coastline	28.573255	-80.646895	28.60283	-80.58815	6.613767
	railway	28.573255	-80.646895	28.57314	-80.65398	0.692172
	highway	28.573255	-80.646895	28.57307	-80.65553	0.843713
	city	28.573255	-80.646895	28.10469	-80.64784	52.118441
VAFB SLC-4E	coastline	34.632834	-120.610745	34.63470	-120.62531	1.349004
	railway	34.632834	-120.610745	34.63585	-120.62401	1.259455
	highway	34.632834	-120.610745	34.70480	-120.56940	8.853368
	city	34.632834	-120.610745	34.64253	-120.47331	12.623666

VAFB SLC-4E | SELECTED MAP POINTS

Railway 1.26 km • Coastline 1.35 km
Highway 8.85 km • City 12.62 km

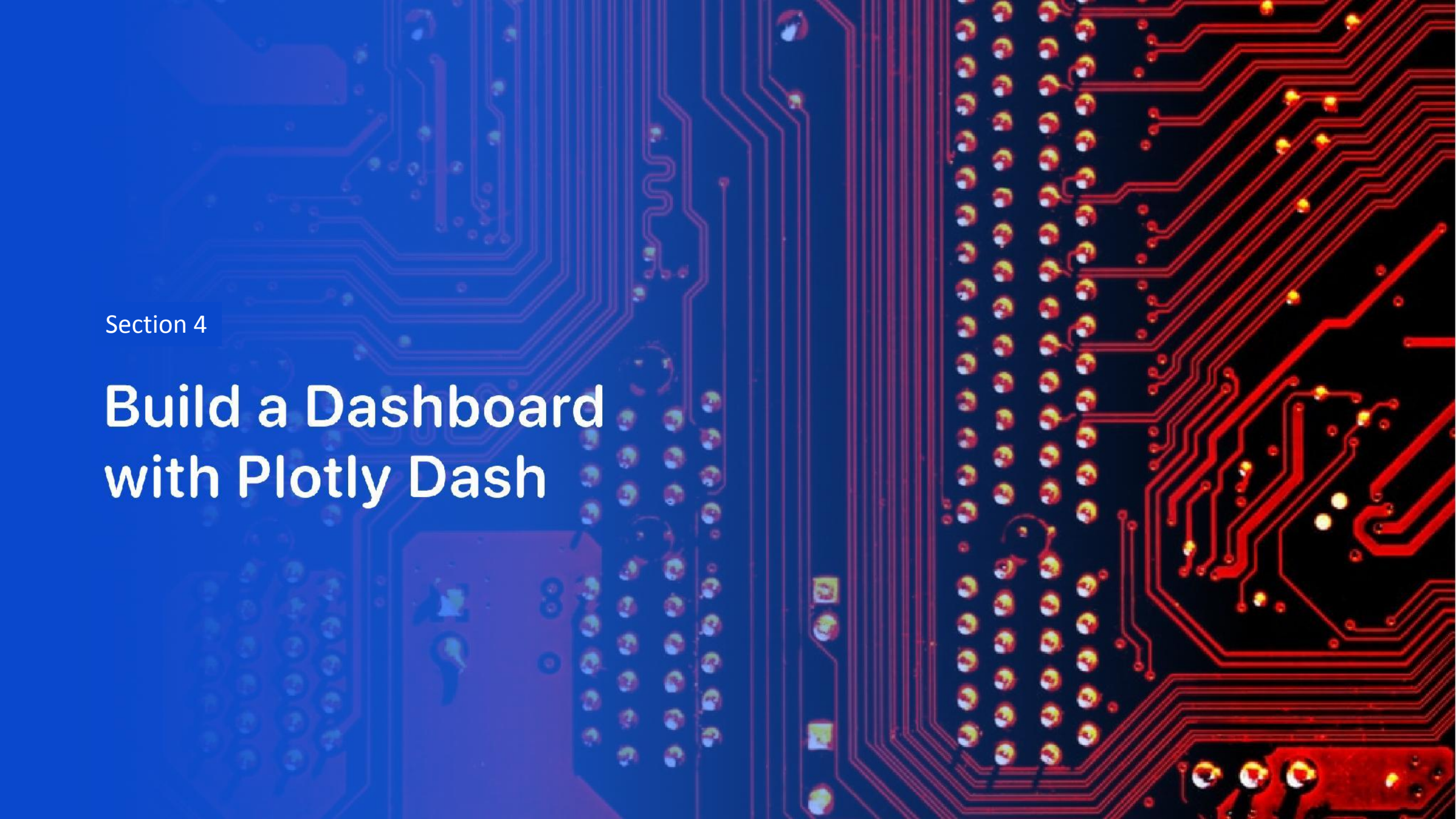
Of these selected points, the railway and coast are closest.

Blue lines connect VAFB to selected feature points. The VAFB rows show the calculated distances. The upper-right coordinates show the cursor position.

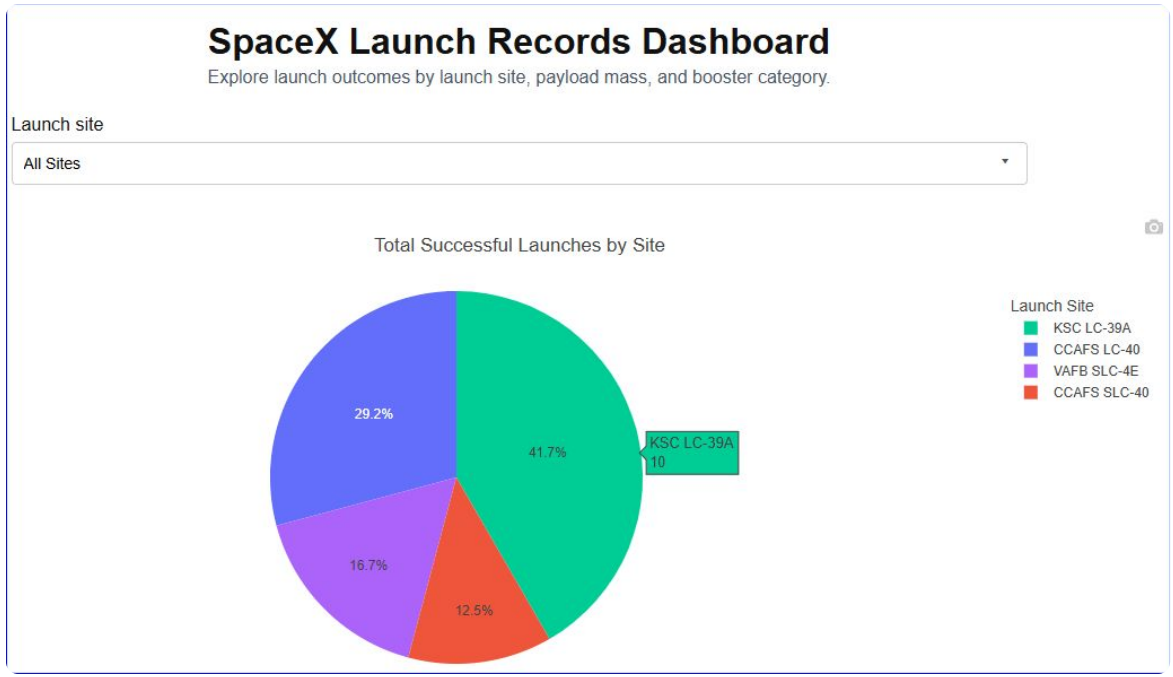
https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%203/lab-jupyter-launch-site-location-v2.ipynb

Section 4

Build a Dashboard with Plotly Dash



Successful Landings by Launch Site



Key Takeaway: Pie slices show landing success share across sites, not individual site launch success rates.

ALL-SITE VIEW

Total Success Distribution

Across four active launch sites

41.7%

KSC LC-39A Share

Contributed 10 of 24 successful landings

TOTAL SUCCESSES

24 Landings

Across all 4 sites

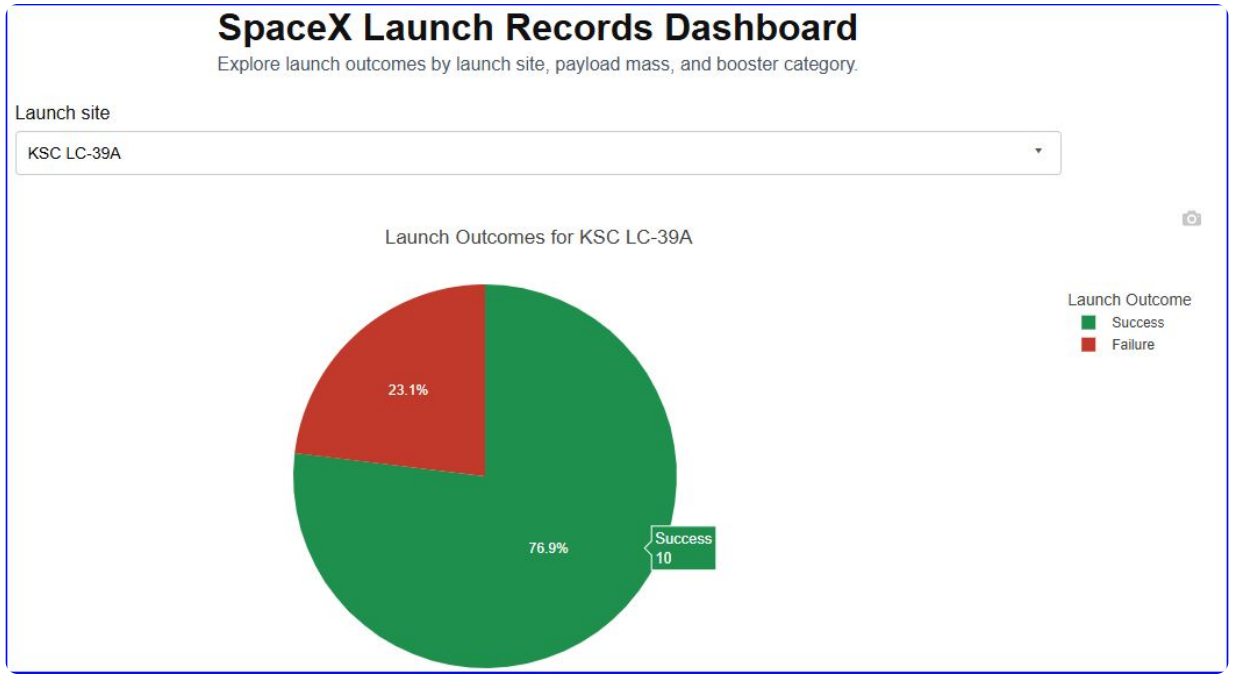
KSC LC-39A (10), VAFB SLC-4E (4), CCAFS LC-40 (7), CCAFS SLC-40 (3).

Live app: https://spacex-launch-analytics.netlify.app/

https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%203/spacex-dash-app.py

SpaceX Launch Analytics • Dashboard Insights

KSC LC-39A: 76.9% Success Rate



Interactive Feature: Selecting KSC LC-39A in the dropdown changes the pie chart from all-site share to KSC's specific launch outcomes.

KSC LC-39A

Site Success Breakdown

Highest success rate in this dataset

76.9%

10 / 13 Successful Landings

Only 3 landing failures out of 13 total attempts

OUTCOME SUMMARY

3 Failures

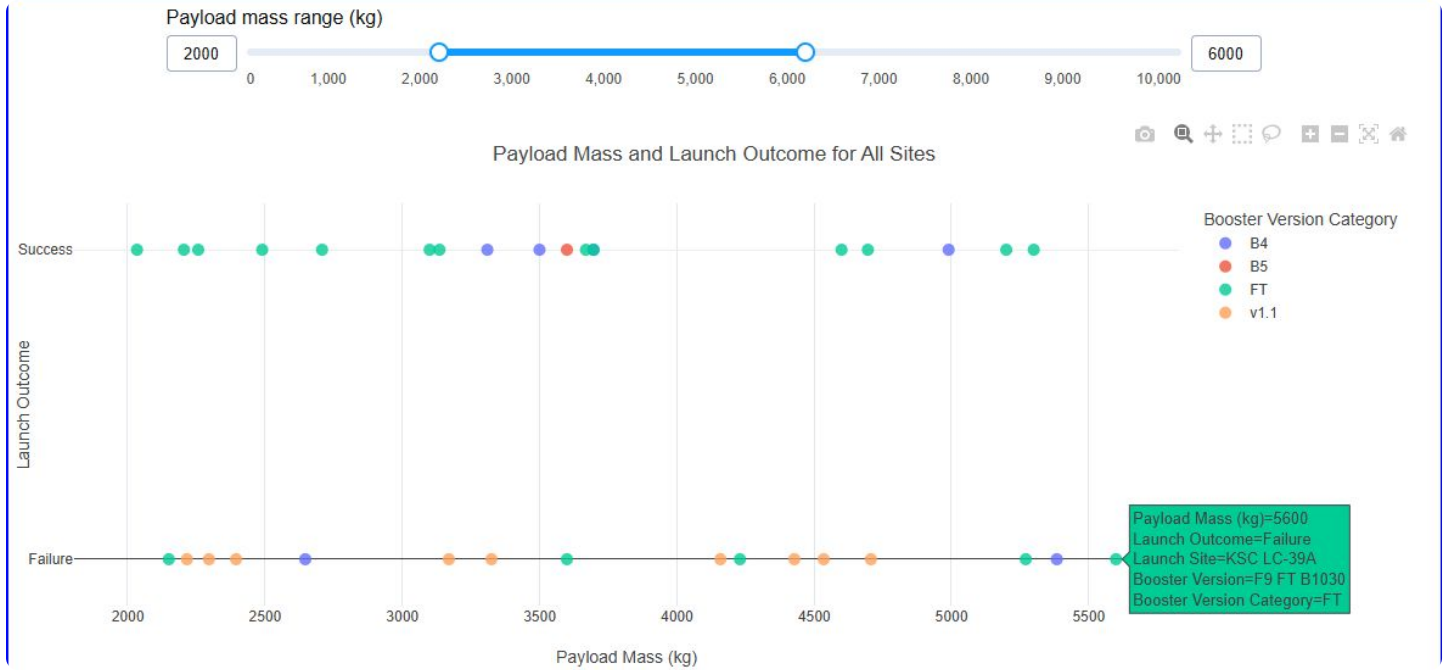
23.1% failure rate

KSC LC-39A demonstrates the highest reliability among all four SpaceX launch facilities.

Live app: https://spacex-launch-analytics.netlify.app/

https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%203/spacex-dash-app.py

Payload Mass vs. Landing Outcome



Key Observation: Each dot represents one launch. Color marks booster category; the hover state details specific failures (e.g., 5,600 kg KSC FT failure).

PAYLOAD FILTER

2,000 – 6,000 kg

Applied across all launch sites

18 / 34

Landings Succeeded

52.9% success rate in this range

FT BOOSTER HIGHLIGHT

13 of 18

Successes in FT category

FT boosters achieve a 72.2% success rate in this specific payload bracket.

Live app: https://spacex-launch-analytics.netlify.app/

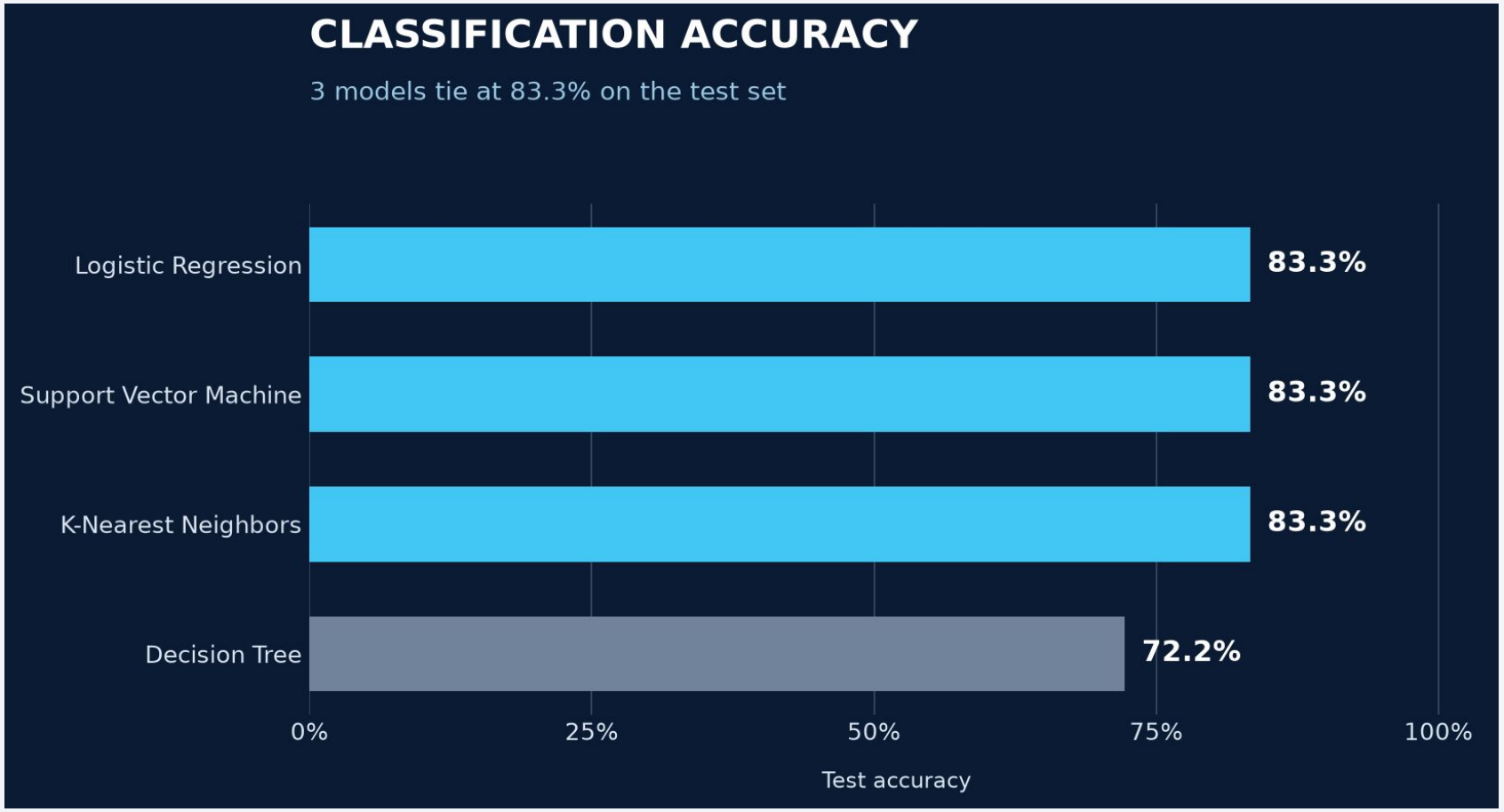
https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%203/spacex-dash-app.py

Section 5

Predictive Analysis (Classification)



Classification Accuracy



TEST SET

83.3%

Three models tie across 18 launches.

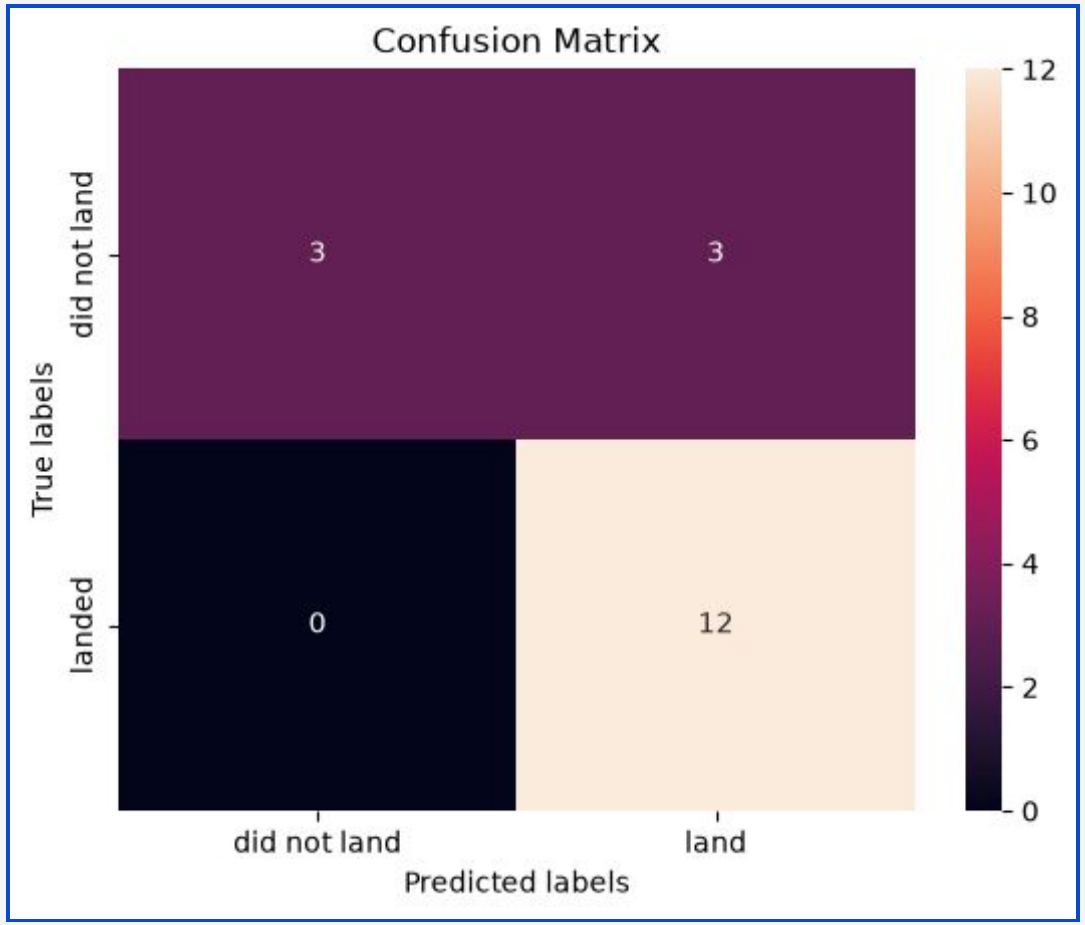
72.2%

Decision tree

SVM selected by 84.8% cross-validation accuracy.

https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%204/SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb

Confusion Matrix




```
Selected model: Support Vector Machine
                  precision    recall  f1-score   support

    Failed Landing       1.00      0.50      0.67         6
Successful Landing       0.80      1.00      0.89        12

          accuracy                           0.83        18
         macro avg       0.90      0.75      0.78        18
      weighted avg       0.87      0.83      0.81        18

True Positive Rate: 100.0%
```

	Model	Accuracy	Precision	True Positive Rate (Recall)	F1 Score	False Positives	False Negatives
0	Logistic Regression	0.833333	0.800000	1.000000	0.888889	3	0
1	Support Vector Machine	0.833333	0.800000	1.000000	0.888889	3	0
2	K-Nearest Neighbors	0.833333	0.800000	1.000000	0.888889	3	0
3	Decision Tree	0.722222	0.769231	0.833333	0.800000	3	2

SVM MODEL | 18 TEST LAUNCHES

15 / 18 correct

12/12 successful landings identified.
3/6 failures caught; the other 3 predicted to land.

https://github.com/treinart/IBM-Data-Science-Professional-Certificate/blob/main/10_Applied%20Data%20Science%20Capstone/Module%204/SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb

Conclusions

By 2019 - 20, landings were common.
The selected SVM still missed half the failures.

LANDING RECORD

9 / 10

Successful landings in 2019

16 / 19

Successful landings in 2020

REUSE AND ORBIT

78.4% vs **58.5%**

Reused (29/37) vs not reused (31/53)

85.7% vs **51.9%**

VLEO (12/14) vs GTO (14/27).
No clear payload mass cutoff.

MODEL TEST

15 / 18

Three models tied at 83.3%.

3 / 6

Failed landings the SVM predicted would land.

NEXT STEPS: Fit scaling within training, test on later flights, and track missed failures alongside accuracy.

Appendix: Standalone Web App

APP.JS: PRECISE PAYLOAD INPUTS

```
function commitPayloadValue(field, slider, otherSlider, isLower) {
  const entered=Number(field.value);
  if(field.value.trim()==='' || !Number.isFinite(entered)) {
    refreshSlider();
    return;
  }
  const bounded=Math.max(0,Math.min(MAX_PAYLOAD,entered));
  slider.value=String(isLower ? Math.min(bounded,Number(otherSlider.value))
    : Math.max(bounded,Number(otherSlider.value)));
  refreshSlider();
  renderScatter();
}
lowValueEl.addEventListener('change',()=>commitPayloadValue(lowValueEl,lowEl,highEl,true));
highValueEl.addEventListener('change',()=>commitPayloadValue(highValueEl,highEl,lowEl,false));
for(const field of [lowValueEl,highValueEl]) {
  field.addEventListener('keydown',event=>{if(event.key==='Enter')field.blur();});
}
```

The function limits entries to 0–10,000 kg and refreshes the scatter plot.

View this function in app.js →

ADDITIONAL BUILD

A browser version of the Python Dash dashboard

HTML, CSS, and JavaScript use the same 56 launch records. Plotly.js redraws the charts when users change the site or payload range. Netlify serves the app as static files.

SOURCE AND DEMO LINKS

Open live dashboard

Read the project README

Browse app.js source

View Python Dash script

View dashboard CSV

IBM Developer
SKILLS NETWORK

IBM APPLIED DATA SCIENCE CAPSTONE

SpaceX Falcon 9 First-Stage Landing Prediction



Thank you!

Travis Reinart

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github.com/treinart/IBM-Data-Science-Professional-Certificate

AVAILABLE FORMATS

PowerPoint Deck PDF Presentation Google Slides